Modern Difference-in-Differences and Synthetic Control

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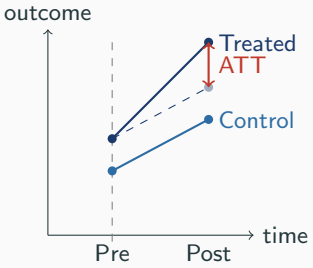
The Limits of TWFE

The Clean Case: DiD 2×2

Two groups, two periods:

- One group treated at $t = 1$, the other never
- The causal effect is a *double difference*

$$\widehat{\delta} = \underbrace{\left(\bar{Y}^{\text{tr}}_{\text{post}} - \bar{Y}^{\text{tr}}_{\text{pre}}\right)}_{\text{change in treated}} - \underbrace{\left(\bar{Y}^{\text{co}}_{\text{post}} - \bar{Y}^{\text{co}}_{\text{pre}}\right)}_{\text{change in control}}$$



The first difference wipes out fixed level gaps. The second wipes out the common time trend.

Key assumption: parallel trends — absent treatment, both groups would have moved *in parallel*. Levels may differ; only the *slopes* must match. This is a claim about a *counterfactual*, so in a 2×2 you cannot test it — you assume it.

From the 2×2 to TWFE

In the 2×2 world, the DiD estimate is the coefficient β_3 in:

$$Y_{it} = \beta_0 + \beta_1 \,\text{Post}_t + \beta_2 \,\text{Treated}_i + \beta_3 \,(\text{Post}_t \times \text{Treated}_i) + \varepsilon_{it}$$

With a panel of many units and periods, researchers run the natural generalisation — two-way fixed effects:

$$Y_{it} = \alpha_i + \lambda_t + \beta \, D_{it} + \varepsilon_{it} \qquad (\alpha_i\text{: unit FE}, \quad \lambda_t\text{: time FE})$$

Here β plays the role of β_3: it is *supposed* to be the ATT. In 2010–2012, 19% of empirical AER articles used it (dCH, 2020).

An assumption that usually stays implicit

If the effect is the same in every (g, t) cell, $\hat{\beta}$ equals it in any design. Otherwise it is a weighted sum of cell effects (weights sum to one, can be negative), equal to the ATT only if weights and effects are uncorrelated. The rest of this block is *when* that matters, and *which toolkit* fixes each case.

First, the Honest Part: When Is TWFE Fine?

TWFE estimates a weighted sum of the cell effects $\Delta_{g,t}$ (cohort g, period t); the weights sum to one but can be negative. Two situations make it trustworthy:

Two routes to safety

- **Homogeneous effects.** If $\Delta_{g,t}$ is the same in every cell, the weighted sum equals it *exactly* — in *any* design.
- **Clean design.** Binary treatment, one adoption date, nobody treated before it: every treated cell gets the same weight. (Two periods alone are not enough.)

Bias, and sign reversal

- **Bias**: effects that **vary** (by cohort or over time), **correlated with the TWFE weights**.
- A **sign flip** also needs **negative weights**, possible with **staggered** timing, **on/off** switching or **non-binary** treatment.

Then a positive effect can enter with a negative weight, and $\hat{\beta}$ shrinks or flips sign.

Homogeneous effects (any design) or a clean design (any heterogeneity): $\hat{\beta}$ is the ATT. Staggered dates, effects differing by cohort: a variance-weighted ATT, plus a bias term if effects change over time.

What TWFE Really Computes: A Weighted Average of 2×2's

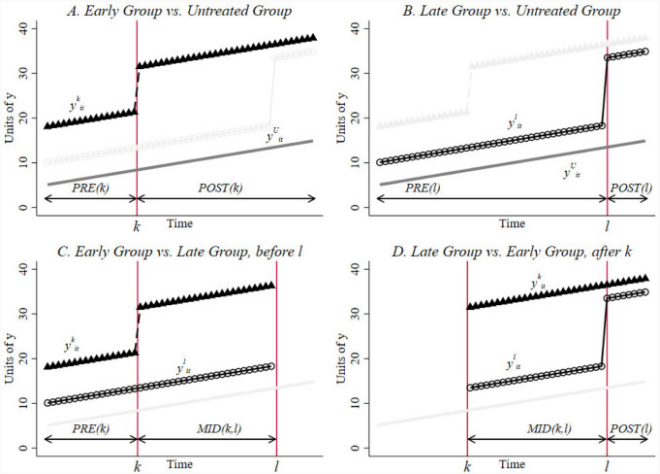
Goodman-Bacon (2021) proves that with binary, staggered treatment:

$$\hat{\beta}^{\text{TWFE}} = \sum_{(k,\ell)} s_{k\ell}\ \hat{\delta}^{2\times 2}_{k\ell}$$

$\hat{\beta}^{\text{TWFE}}$ is a weighted average of every possible 2×2 DiD between two timing groups (the never-treated included). Each 2×2 compares average outcomes over two time windows set by the adoption dates (before, between and after adoption), not two single dates.

What matters is which comparisons enter that average. Some are clean. One compares a newly-treated group to a group *that is already treated* — and *when effects change over time*, that comparison is what biases $\hat{\beta}$.

The Four Comparisons Behind $\hat{\beta}^{\text{TWFE}}$



A, B — early/late treated vs. *never-treated*. Clean controls.

C — early vs. late, *before* the late group is treated. The late group is still untreated, so also a clean control.

D — late vs. early, *after* the early group is treated. Now the "control" already has the treatment. This is the forbidden comparison.

Goodman-Bacon (2021), J. Econometrics, Fig. 2. Dark series: the data each 2×2 uses; faded grey: left out.

Failure #1: The Forbidden Comparison

→ Block 1

Think of a reform rolled out in waves — say no-fault divorce laws, adopted by early states and, years later, by late states.

To estimate the *late* states' effect, TWFE needs a control for them. Panel D used the early states — but by then they have had the law for years. Their outcome already reflects the treatment.

They are not a "no-law" counterfactual. If the early states' effect kept growing, TWFE subtracts that accumulated change from the late states' estimate.

Consequence
The late effect looks too small — or flips negative. Not because the law hurt anyone, but because the control was contaminated by its own treatment.

The fix (Block 1): never use an already-treated unit as a control. Build clean cohort-by-cohort comparisons — the **Callaway & Sant'Anna** idea.

Failure #2: Where Negative Weights Come From

Start from what any DiD does: compare a treated unit to a control.

$$\text{ATT} = \sum \Big(\underbrace{w_+ \, Y_{\text{treated},\,t}}_{\text{added}} - \underbrace{w_- \, Y_{\text{control},\,t}}_{\text{subtracted}} \Big)$$

The treated unit enters with a positive weight; the control enters *subtracting* — an effective negative weight. Normally this is fine: the control is *untreated*, so the negative weight sits on an untreated unit.

The staggered twist

With staggered timing, TWFE uses an already-treated unit as the control for a later one:

$$\text{ATT} = \sum \Big(w_+ \, Y_{\text{later},\,t} \; - \; w_- \, Y_{\text{earlier-treated},\,t} \Big).$$

Now the negative weight sits on a treated unit — whose own effect should count *positively*.

That misplaced sign is the whole story: TWFE puts negative weight on some *treated* units, because it presses them into service as controls.

Failure #2: . . . and It Can Flip the Sign

→ Block 2

When does the misplaced weight do damage? It depends on the earlier unit's effect.

Constant effect: harmless
A one-time, constant shift leaves the earlier unit's *trend* parallel to the later one's. It is still a usable control, and the negative weight does no harm.

Changing effect: damage
If the earlier unit's effect *grows or fades*, its post-treatment trend is no longer parallel. Using it as a control is invalid, and the negative weight distorts the total.

de Chaisemartin & D'Haultfœuille (2020), the example on pp. 2970–71
Two groups, three periods. Group 2 is treated in periods 2 and 3, group 1 in period 3 only. TWFE returns $\beta_{fe} = \frac{1}{2}\,\Delta_{1,3} + \Delta_{2,2} - \frac{1}{2}\,\Delta_{2,3}$: the cell where group 2 serves as the control for group 1 enters with weight $-\frac{1}{2}$. With cell effects $(1, 1, 4)$, $\beta_{fe} = \frac{1}{2} + 1 - 2 = -\frac{1}{2}$: every effect is positive and the estimate is negative.

Binary staggered designs are exposed; continuous and switching ones more so. **The fix: controls still at the unit's baseline treatment (binary: never already-treated).**

It Is Not Only Timing: Who Responds

With one common date and only unit and period FE, TWFE is the 2×2 DiD and returns the ATT: every treated unit counts equally. Add urban/rural-by-period FE and it weights each subgroup by its size times the variance of its treatment.

With those FE, subgroups of similar size:

- Urban: strong effect, but *most* units treated ⇒ little variation ⇒ *low* weight
- Rural: weak effect, but an even *split* treated/untreated ⇒ *high* weight

Size × variance weight, not population weight

$\hat{\beta} = \sum_g w_g \beta_g$, with $w_g \propto n_g\, p_g(1 - p_g)$ (p_g: treated share): size times treatment variance, not effect size. Here it over-weights the weakly-affected rural group.

No sign reversal here — all weights are positive. But the estimate is a variance-weighted ATT, not the simple average effect. The subgroup-by-period FE create these weights; a time-invariant urban dummy is absorbed by the unit FE and changes nothing.

And You Cannot Even Diagnose It: The Pre-Trend Trap

You run an event-study plot. Pre-treatment coefficients look flat. You conclude "parallel trends holds, TWFE is valid."

Why that reasoning fails

In a staggered TWFE event study, each pre-period coefficient mixes in cohorts' effects from other relative periods, which cancel only if effects are homogeneous across cohorts (Sun & Abraham, 2021). The pre-trend can look flat when parallel trends is violated — and non-flat when it holds.

- Passing a pre-trend test $\neq$ parallel trends holds
- Failing a pre-trend test $\neq$ parallel trends is violated

Under staggered adoption the TWFE event study is unreliable *in both directions* — so the usual "check" gives false comfort. Blocks 1 and 2 give event studies free of this contamination; a flat pre-trend there still does not prove parallel trends (pre-tests have low power).

Failure #3: Too Few Treated Units

→ Block 3

Sometimes the problem is not the estimator — it is that you have one treated unit (or just a handful): a single country, a single state, one reform.

With $N_{\text{treated}} = 1$:

- Often no single untreated unit is a good comparison — why this neighbour and not that one?
- Parallel trends against the donor average often fails before treatment (Abadie, 2021, Fig. 1A)
- Cluster / asymptotic inference breaks — you cannot appeal to "many treated clusters"

A different tradition
Synthetic control builds the counterfactual as a weighted blend of untreated units that tracks the treated unit before treatment. Identification is *pre-period fit under a factor model*; inference is typically by *permutation* (placebo units), not asymptotics.
The fix (Block 3): SCM and SDID.

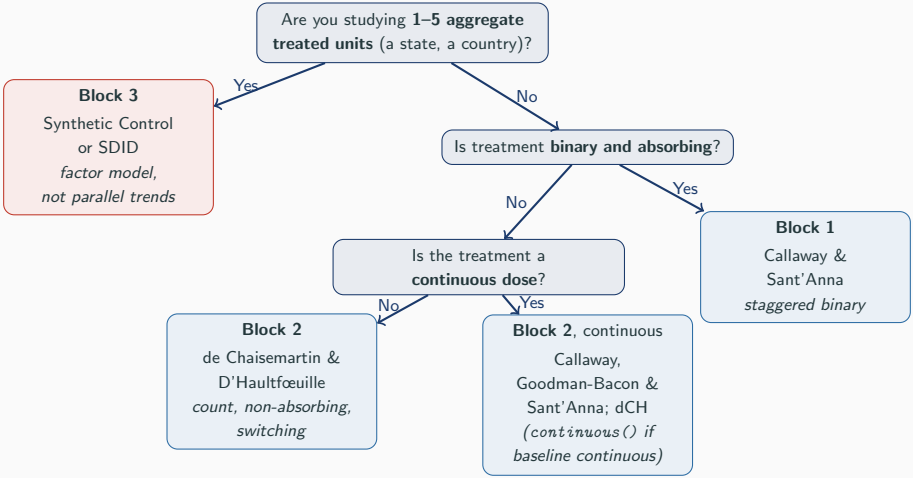
What Broke, in One View

1. **Wrong control** — staggering makes TWFE use already-treated units as controls. → Block 1
2. **Wrong sign / wrong weights** — heterogeneous and dynamic effects get mechanical, sometimes negative, weights. → Block 2
3. **Too few treated units** — no single good control, parallel trends often fails pre-treatment, no sampling-based inference. → Block 3

Tradition 1 — heterogeneity-robust DiD
Blocks 1 & 2. Keep parallel trends and many treated units; fix the weighting by building clean comparisons.

Tradition 2 — comparative case studies
Block 3. A few aggregate treated units; build the counterfactual under a factor model, not parallel trends.

Which Toolkit for Your Study?



Binary absorbing: dCH equals Callaway & Sant'Anna with not-yet-treated controls, not an independent check.

Course Roadmap

Four blocks

- **Block 0**: why TWFE fails, and the map
- **Block 1**: Callaway & Sant'Anna
- **Block 2**: de Chaisemartin & D'Haultfœuille, then continuous treatment (Callaway, Goodman-Bacon & Sant'Anna)
- **Block 3**: Synthetic Control + SDID

Each methods block: theory → applied cases with real data → Stata demo.

By the end you can:

- Diagnose whether a TWFE estimate is trustworthy
- Pick the right estimator for your data structure
- Run it in Stata (`csdid`, `did_multiplegt_dyn`, `sdid`)
- Read and report the output correctly

Every figure from Block 1 onward comes from Stata or R code shown on the slides, run on published data (one simulated panel, flagged on its slide). The Block 0 comparison diagram is reproduced from Goodman-Bacon (2021).

Callaway & Sant'Anna (2021)

Where We Left Off: The TWFE Problem

Block 0 showed: TWFE with staggered adoption is a weighted average of all possible 2×2 DiDs, and the weights on those 2×2 *blocks* are all positive (Goodman-Bacon).

(Block 2 revisits the same designs through dCH's cell-level weights, where some already-treated cells get negative weight — the central issue once treatment is non-binary or non-absorbing.)

The root cause: TWFE uses already-treated units as controls for later-treated units; if their effects change over time, that change is subtracted and biases $\hat{\beta}$.

The fix requires three things:

1. Build every comparison using only *clean* control units
2. Allow treatment effects to differ across groups and over time
3. Aggregate the results into summaries we can interpret

This block: Callaway & Sant'Anna (2021) deliver all three.
Use case: binary, absorbing, staggered treatment with many treated units.

The Key Idea: Think in Cohorts

Instead of one $\hat{\beta}$, estimate one ATT per (cohort, calendar year) cell.

Define:

- g = the year a unit is *first* treated (its "cohort")
- t = the calendar year of measurement

Group-Time ATT

$$\text{ATT}(g,t) = E[\, Y_t(g) - Y_t(0) \mid G = g \,]$$

Average treatment effect for cohort g at time t, averaged only over units in that cohort.

A grid, not a single number:

- One $\text{ATT}(g,t)$ per (cohort, period) cell, $t \geq 2 \Rightarrow$ a $G \times (T-1)$ grid
- Cells with $t < g$: *pre-treatment*, ≈ 0 under no anticipation and parallel trends
- Cells with $t \geq g$ are the effects of interest

No restriction imposed
Effects may differ *across cohorts* and *over time*. TWFE forces them into one $\hat{\beta}$; C&S keeps them separate.

The ATT(g, t) Grid, Made Concrete

Each box is one cell ATT(g, t): row = cohort (year first treated), column = calendar year. Three cohorts plus a never-treated group:

	2004	2005	2006	2007
treated 2004	ATT	ATT	ATT	ATT
treated 2006	pre	pre	ATT	ATT
treated 2007	pre	pre	pre	ATT
never treated	–	–	–	–

ATT effect ($t \geq g$) pre pre-treatment check ($t < g$, ≈ 0) – never-treated control

Assumption 1: Staggered (Absorbing) Treatment

Once treated, always treated:

$$D_{i1} = 0 \quad \text{and} \quad D_{i,t-1} = 1 \;\Rightarrow\; D_{it} = 1 \quad (t = 2, \ldots, T)$$

Nobody is treated in period 1 (such units are dropped); $G_i = g$ if first treated in g.

Why this matters:

- Each unit belongs to exactly one cohort
- Pre-treatment periods of a cohort are clean
- Placebo "pseudo" ATT(g, t) for $2 \leq t < g$: ≈ 0 under PT and no anticipation

Common examples:

- Minimum wage laws (state adoption)
- ACA Medicaid expansion
- Environmental regulations
- No-fault divorce laws

Not satisfied: treatment that can be reversed or is continuous. $\rightarrow$ Block 2.

Assumption 2: Parallel Trends (C&S Assumptions 4–5)

Unconditional version (simplest), for each cohort g and each $t \geq g$:

$$E[Y_t(0) - Y_{t-1}(0) \mid G = g] = E[Y_t(0) - Y_{t-1}(0) \mid \text{control group}]$$

Absent treatment, cohort g moves in parallel with its controls from $g-1$ on.

Identification uses the **long difference** *relative to $g-1$ (the cohort's last pre-treatment year), not period-by-period first differences — so $g-1$ is the natural baseline (relative time $e = -1$).*

Conditional version (more flexible), for each g and $t \geq g$:

$$E[Y_t(0) - Y_{t-1}(0) \mid X,\ G = g] = E[Y_t(0) - Y_{t-1}(0) \mid X, \text{ control group}]$$

Use it when pre-treatment X that drive the trend of $Y(0)$ differ across groups.

Why this matters
The conditional version is what motivates the doubly-robust estimator.

Who Is the Control Group?

C&S offers two clean options. The choice matters for both credibility and power.

Never-treated units
Units that are never treated throughout the sample.

Pro: most stable counterfactual
Con: may be too few; may differ systematically

Not-yet-treated units
Units that will eventually be treated, but have not been treated *yet* at time t.

Pro: more comparison units; arguably more similar
Con: imposes restrictions on pre-trends across cohorts

What is excluded in both cases: already-treated units — the source of TWFE bias.

The Doubly Robust Estimator

For each (g, t) cell, C&S identifies ATT using one of three equivalent approaches:

1. **Outcome regression (OR):** model the controls' change $E[Y_t - Y_{g-1} \mid X]$ and impute it for cohort g
2. **IPW:** re-weight control units by the odds $p_g(X)/(1 - p_g(X))$ of being in cohort g rather than control
3. **Doubly robust (DR):** combine both

Why doubly robust?
Consistent if *either* the propensity-score model *or* the outcome regression is correctly specified — not necessarily both.

In practice: the doubly-robust estimator is the default — and the authors' recommendation.

Required: overlap. $P(G_g{=}1) > \varepsilon$ and $p_{g,t}(X) < 1{-}\varepsilon$ for almost every X. As $p \to 1$, IPW weights $p/(1{-}p)$ blow up and IPW/DR become very imprecise.

From ATT(g, t) to Summaries: Aggregation

With G cohorts and T periods, you have up to $G \times (T-1)$ estimates (period 1 is only a base period). Aggregate them into one of four summaries:

Summary	**Parameter**	**Use when...**
Event study	$\theta_{es}(e)$	Show dynamics; assess pre-trends
Group-specific	$\theta_{sel}(g)$	Compare early vs. late adopters
Calendar time	$\theta_c(t)$	Effects in a specific period
Overall ATT	θ_O	Single number for abstract/table

Key: all four are weighted averages of the same ATT(g, t) grid — they answer different questions, not different models.

Aggregation: The Formal Estimands

All four summaries are weighted averages of the $\mathrm{ATT}(g,t)$ grid (Callaway & Sant'Anna 2021):

$$\theta_{es}(e) = \sum_g \mathbf{1}\{g{+}e \le T\}\, P(G{=}g \mid G{+}e \le T)\, \mathrm{ATT}(g, g{+}e) \qquad \text{event study}$$

$$\theta_{sel}(g) = \tfrac{1}{T-g+1} \sum_{t \ge g} \mathrm{ATT}(g,t) \qquad \text{group-specific}$$

$$\theta_{c}(t) = \sum_g \mathbf{1}\{t \ge g\}\, P(G{=}g \mid G \le t)\, \mathrm{ATT}(g,t) \qquad \text{calendar}$$

$$\theta_{O} = \sum_g \theta_{sel}(g)\, P(G{=}g \mid G \le T) \qquad \text{overall}$$

Two ways to weight the "overall" effect

θ_O^{sel} averages *within* groups first, then across groups — the authors' preferred summary. θ_O^{W} weights every post-treatment cell by group size, putting more weight on early-treated groups. In general they differ; they coincide when every cohort has the same $\theta_{sel}(g)$ (for example, under homogeneous effects).

Event Study $\theta_{es}(e)$: Effect by Time Since Treatment

Question How does the effect evolve with exposure length $e = t - g$?

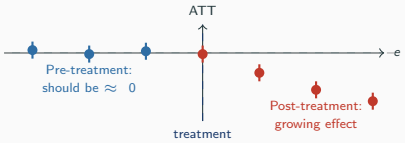
Averages the cells $\text{ATT}(g, g+e)$ at horizon e, weighted by cohort size. $e{=}0$ is the on-impact effect; $e{<}0$ is pre-treatment.

Use it the headline plot — trace whether effects build, fade, or persist.

Gotcha the $e{<}0$ points are *not* effects: they are the parallel-trends check and should sit near 0. Read the whole path, not a single horizon.

The Event Study Plot

Most useful output from C&S: the dynamic aggregation $\theta_{es}(e)$ — the average effect e periods after first treatment.



Confidence intervals: use **simultaneous** (joint) bands — wider than pointwise, correct for reading the whole pre-trend path as one joint test.

Reading the plot:

- $e < 0$: pre-treatment. Should be ≈ 0 — an honest pre-trends check (but low power, and conditioning on passing it distorts inference, Roth 2022).
- $e \geq 0$: post-treatment effects. Slope shows whether the effect grows, fades, or is constant.

Unlike TWFE: pre-trends here are not contaminated by heterogeneous post-treatment effects.

Group Effect $\theta_{sel}(g)$: One Cohort's Average

Question What was the average effect for the cohort first treated at g, over its post-treatment life?

Averages $\text{ATT}(g,t)$ for $t \geq g$, with *equal* weight $1/(T-g+1)$ — a single cohort, so no group-size weighting.

Use it compare **early vs. late adopters**: is the effect larger for those treated sooner?

Gotcha later cohorts have fewer post-periods, so their average covers a *shorter* horizon — not a like-for-like comparison with early cohorts.

Calendar Effect $\theta_c(t)$: One Year's Average

Question What was the average effect in calendar year t, across everyone treated by then?

Averages $\text{ATT}(g, t)$ over cohorts with $g \leq t$, weighted by cohort size $P(G{=}g \mid G \leq t)$.

Use it when a specific **period** matters — a recession, a national shock — and you want the effect in *that* year.

Gotcha a later year is not "longer exposure": it has *more* cohorts treated. Moving across t changes who is included, not just the dynamics.

Overall Effect θ_O: The Single Number

Question One headline number for the abstract or a summary table.

Averages each cohort's own average $\theta_{sel}(g)$ first, then across cohorts weighted by size $P(G{=}g \mid G \leq T)$ — the 2×2 ATT analog.

Use it the scalar to report *alongside* — never instead of — the event study.

Gotcha it hides all dynamics and cohort heterogeneity: a single -0.03 can mask effects that grow, fade, or differ sharply by cohort.

Assumption 3: No Anticipation (C&S Assumption 3, $\delta = 0$)

Units in cohort g behave as untreated in all periods before g:

$$E[Y_t(g) \mid G = g] = E[Y_t(0) \mid G = g] \quad \text{for all } t < g$$

In practice, this means:

- Units did not change behavior before the treatment date in anticipation of being treated
- Pre-treatment ATT(g, t) are zero under this assumption; nonzero pre-period estimates signal a failure of it or of parallel trends

When might it fail?

- Firms adjust before a law takes effect
- States change spending in anticipation of a federal matching program

Partial fix: C&S allow a *known* anticipation window $\delta \geq 0$ (e.g., 1 year): the reference period moves to $g - \delta - 1$, and parallel trends must then hold over more periods.

When Is C&S the Right Choice?

Use C&S when:

- Treatment is **binary and absorbing**
- Many treated units (across multiple cohorts)
- You suspect treatment effect **heterogeneity** across cohorts or dynamics over time
- You want an honest pre-trends test
- Some groups differ on observables → use covariates with the DR estimator

Use something else when:

- Treatment is **non-absorbing / continuous** → Block 2 (de Chaisemartin & D'H; for a continuous dose, Callaway, Goodman-Bacon & Sant'Anna)
- Very few treated units → Block 3 (Synthetic Control)
- You are *certain* treatment effects are homogeneous and constant → TWFE recovers the ATT

For the mathematically inclined

Formal identification results and asymptotic theory: Callaway & Sant'Anna (2021), *Journal of Econometrics* 225(2), pp. 200–230. Doubly-robust estimand derived in Sant'Anna & Zhao (2020).

Applications

Applied Case: Real Data, Reproducible Stata

One estimator, the full C&S workflow on real data.

Minimum wage and teen employment (Callaway & Sant'Anna 2021)
Data: `mpdta` — 500 US counties × 2003–2007

Outcome: log teen employment (`lemp`)

Treatment: state minimum-wage increase; 3 cohorts (2004, 2006, 2007) + never-treated

Why C&S: staggered timing across cohorts, with effects that may differ by cohort and grow with exposure — exactly where TWFE's forbidden comparisons bite.

Roadmap: estimate the ATT(g,t) grid, then read it through each aggregation — event study $\theta_{es}(e)$, by cohort $\theta_{sel}(g)$, by calendar year $\theta_c(t)$, and the overall θ_O.

Minimum Wage and Teen Employment

Research question: did state minimum wage laws reduce teen employment?

Why C&S, not TWFE?

- State minimum wage laws adopted in different years — three cohorts: 2004, 2006, 2007
- TWFE uses already-treated counties as controls for later-treated counties — the forbidden comparison
- If employment effects grow with exposure time, late cohorts' controls are contaminated by early-cohort post-treatment effects

Data structure:

- `mpdta`: 500 counties $\times$ 5 years (2003–2007), 2,500 obs
- Outcome: `lemp` — log teen employment per county
- Cohort variable: `first_treat` $\in \{2004, 2006, 2007, 0\}$ ($0 =$ never treated)

Stata: Estimation, Aggregations, TWFE Benchmark

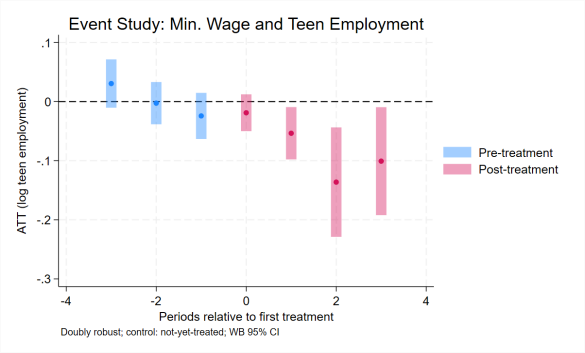
```
* ssc install csdid drdid, replace

use "$datadir/mpdta.dta", clear
xtset county_id year

* Step 1: estimate all ATT(g,t) cells (doubly robust) and save the RIF file
csdid lemp, ivar(county_id) time(year) gvar(first_treat) ///
    method(dripw) notyet saverif(rif_mpdta) replace
* Step 2: aggregations from the saved RIF file -- each is one theta
preserve
use rif_mpdta, clear
csdid_stats simple,   wboot rseed(1234) /* theta_O^W    = -0.040  */
csdid_stats group,    wboot rseed(1234) /* theta_sel(g); theta_O = -0.030 */
csdid_stats event,    wboot rseed(1234) /* theta_es(e): dynamic   */
csdid_stats calendar, wboot rseed(1234) /* theta_c(t):  by year   */
* Step 3: event-study plot
qui csdid_stats event, wboot rseed(1234)
csdid_plot, title("Event Study: Min. Wage and Teen Employment") ///
    xtitle("Periods relative to first treatment")               ///
    ytitle("ATT (log teen employment)")
graph export "$figdir/mpdta_eventstudy.png", replace width(1200)
restore
* Static TWFE benchmark (clustered by county), on mpdta
gen byte treat_post = (first_treat>0 & year>=first_treat)
xtreg lemp treat_post i.year, fe vce(cluster county_id) /* -0.037* */
```

Key: `method(dripw)` = doubly robust; `notyet` = not-yet-treated controls; `saverif()` = RIF file read by `csdid_stats`; `wboot` = wild bootstrap SEs.

Event Study: Dynamic Effects $\theta_{es}(e)$



Estimand: $\theta_{es}(e)$ (dynamic).

Pre-trends ($e < 0$): Pre_avg $= 0.001$ (ns); $e = -3$: 0.031 ($p = 0.04$). No evidence against PT on average (not proof).

Post-treatment: grows to $\theta_{es}(2) = -0.136$ log points (2004 only); Post_avg $= -0.077$.

Overall (θ_O^{sel}): -0.030 (SE 0.012, sig at 5%) — a $\sim$3% employment drop.

vs. static TWFE: -0.037 (sig) on `mpdta` — close here. The larger gap (TWFE -0.8% ns vs DR with covariates -3.1%) is in the paper's full sample (C&S 2021, Table 3b).

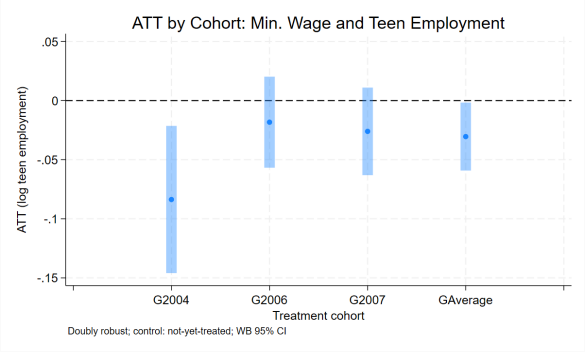
Stata: ATT by Cohort $\theta_{sel}(g)$

```
* Same estimated model (saved RIF file) -- aggregate by cohort
use rif_mpdta, clear
csdid_stats group, wboot rseed(1234)

csdid_plot, title("ATT by Cohort: Min. Wage and Teen Employment") ///
    xtitle("Treatment cohort")                                    ///
    ytitle("ATT (log teen employment)")
graph export "$figdir/mpdta_group.png", replace width(1200)
```

What changes: `group` instead of `event`. Same model, different aggregation of the ATT(g, t) grid.

Effect by Cohort: $\theta_{sel}(g)$



Estimand: $\theta_{sel}(g)$ — average effect for each cohort over its post-treatment years; GAverage $= \theta_O^{sel} = -0.030$.

Cohort 2004: ATT ≈ -0.08 (significant) — the largest effect. Early adopters faced higher minimum wages for more years within the sample window.

Cohorts 2006, 2007: small and not significant — less post-treatment exposure.

Key lesson: heterogeneity across cohorts is real. Pooling them into a single TWFE $\hat{\beta}$ hides it and can bias it: TWFE subtracts the growth in cohort 2004's effect when 2004 controls for later ones.

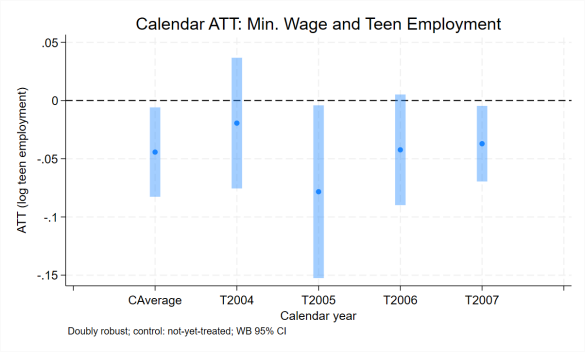
Stata: ATT by Calendar Year $\theta_c(t)$

```
* Same estimated model (saved RIF file) -- aggregate by year
use rif_mpdta, clear
csdid_stats calendar, wboot rseed(1234)

csdid_plot, title("Calendar ATT: Min. Wage and Teen Employment") ///
    xtitle("Calendar year")                                       ///
    ytitle("ATT (log teen employment)")
graph export "$figdir/mpdta_calendar.png", replace width(1200)
```

What changes: `calendar` instead of `event`. Same model, different aggregation of the ATT(g, t) grid.

Effect by Calendar Year: $\theta_c(t)$



Estimand: $\theta_c(t)$ — average effect across cohorts already treated by calendar year t (CAverage $= -0.044$).

2004–2005: only the 2004 cohort is treated; its effect builds ($-0.02 \to -0.08$).

2006–2007: the 2006 and 2007 cohorts (40 and 131 counties, against 20 in 2004) enter with smaller effects and pull the average toward -0.04.

Use it to ask: "what was the average effect across all treated units in year t?" — aggregate policy impact in a given year.

C&S in Practice: Checklist

Before you run the code:

1. Confirm treatment is **binary and absorbing**
2. Decide on comparison group: never-treated (the default) or not-yet-treated (`notyet`)
3. Decide on estimator: doubly robust, `method(dripw)`, the default in `csdid` and in R's `did`
4. If groups differ on observables: add covariates to the model
5. If few treated clusters: SEs and `wboot` unreliable (Roth et al. 2023, Sec. 5.1)

After you run the code:

1. **Inspect ATT(g,t) grid:** do pre-treatment cells look like zero?
2. **Plot the event study:** pre-periods near zero? Not proof of PT: report HonestDiD $\bar{M}$.
3. **Check cohort-specific effects:** is there heterogeneity across cohorts? If yes, report it — do not just report the overall ATT.
4. **Compare to TWFE:** they can be close, but report C&S either way.

Stata: `csdid` + `drdid` from SSC.

de Chaisemartin & D'Haultfœuille (2020 & 2024)

What Callaway & Sant'Anna Assume

Callaway & Sant'Anna handle the staggered binary case: an absorbing 0/1 treatment, identified from never- and not-yet-treated controls.

However, many designs do not fit that framework:

- **Non-binary treatment.** The treatment takes several values (a count, an intensity, a continuous dose), so "treated versus untreated" is no longer the relevant contrast.
- **Non-absorbing treatment.** The treatment can switch on and off, or rise and fall, so there is no clear and unchanged date at which a unit becomes treated

The Non-Binary Problem: Who Is the Control?

When the treatment takes several values, "treated versus untreated" stops being a well-defined contrast.

Three counties, one change

In 2010 county B goes from 2 to 4 branching laws. County A has had 0 all along; county C has had 2 all along.

The change to subtract from B's is C's: C starts where B started, so $\Delta Y_B - \Delta Y_C$ isolates the two extra laws.

A is the wrong benchmark: a county that never had a law can sit on a different trend from counties that have lived with two for years.

What a usable comparison needs

- A control that shares the treated unit's *starting point*, so the only thing separating them is the change itself.
- "Untreated" cannot deliver that when the treatment is a count or an intensity. There may be no untreated unit anywhere in the panel.

Why C&S does not solve it

It is built for a binary indicator with one first-treatment date per unit. A dose enters only binarized, which discards it, as C&S note.

dCH answers by matching on the *baseline dose*: the control for a group leaving $D = 2$ is another group still sitting at $D = 2$. That is the subject of the next few slides.

The Non-Absorbing Problem: No Cohorts

C&S assumes treatment is permanent once adopted ($D_{i,t-1} = 1 \Rightarrow D_{i,t} = 1$). When treatment can switch off again, that assumption fails and the cohort idea stops working.

With reversible treatment the whole path matters: a unit treated in some periods and untreated in others has no single first-treatment date, so the C&S cohort variable is undefined.

TWFE does not solve it either. It assumes the outcome depends only on the current treatment, so once past treatment still affects the outcome the estimate is biased, even when the treatment is binary.

The dCH approach
Date each group by its first treatment change F_g (on or off, up or down) and compare its outcome change since $F_g - 1$ with groups that had the same period-1 treatment and have not changed yet.

Two dCH Papers: Static (2020) vs. Dynamic (2024)

"dCH" is really two papers. Know which one you are using.

dCH (2020, AER)

Static: the *instantaneous* effect, in the period treatment changes.

The potential outcome is $Y_{g,t}(d)$, a *single* value. That is where no-carryover sits: not a separate assumption to look up, but the consequence of indexing by one number instead of by a path.

dCH (2024, REStat)

Dynamic, the version we use: $Y_{g,t}(d_1, \ldots, d_t)$, indexed by the whole treatment path.

The estimand is the actual outcome against having kept the baseline dose throughout, so it **accumulates** with ℓ.

Why lagged effects matter:

- A hurricane hits once, and the county is back at $D = 0$ the next year.
- Under the 2020 notation that county is indistinguishable from one never hit, so it can serve as a *control* for a county hit that year.
- If the damage persists, the control is contaminated and the comparison is biased.

The same design, later today

Deryugina (Case 2) is exactly this: binary and non-absorbing, with estimated effects that keep rising from the hurricane year to five years later.

Set-Up: Groups, Periods, and Treatment

The framework: G groups observed over T periods, with outcome $Y_{g,t}$ and treatment $D_{g,t} \geq 0$.

What is a "group" g?
The unit whose outcome you follow, with *one* treatment value per group and period. It can be either:

- a single **individual unit** (a firm, a person) tracked over time, **or**
- a **group of individuals** (a county, a state), then $Y_{g,t}$ is the group's average outcome.

The mechanics are the same either way.

Treatment assigned above the unit:

- State law, county data: g = county or state; with g = state, a state weighs its number of counties.
- g = county, `cluster(state)`: clusters can be coarser than g, not finer.

Baseline treatment $D_{g,1}$
The treatment level in the *first observed period*. It need **not** be zero, and groups are compared only to others with the *same* $D_{g,1}$.

Starting treated is fine if pre-sample treatment has no lasting effect; if it lasts $\leq k$ periods, start at $k{+}1$ with groups unchanged so far (WA §1.8).

The Timing of a Switch: F_g, Baseline, and Horizon ℓ

Everything is dated relative to each group's own first switch.

Three dates

- $F_g = \min\{t \geq 2 : D_{g,t} \neq D_{g,t-1}\}$: the **first period treatment changes**.
- $F_g - 1$: the **baseline** (last pre-switch period).
- $F_g - 1 + \ell$: the period ℓ **horizons after** the switch, where the effect at horizon ℓ is measured.

How far can ℓ go? (T_g)
T_g = last period for which a same-baseline group still exists that has *not* switched. Group g enters the event study only up to $\ell = T_g - F_g + 1$.

Example: a group switching $0 \to 2$ at $F_g = 4$

t	1	2	3	4	5	6
$D_{g,t}$	0	0	0	2	2	2
role			base	ℓ=1	ℓ=2	ℓ=3

Baseline $= F_g - 1 = 3$; switch at $F_g = 4$.

The effect at horizon ℓ compares the outcome at $F_g - 1 + \ell$ to the baseline, netting out what same-baseline not-yet-switched groups did over the same window.

The dCH Solution: Switchers and Stayers

Core idea: date each group by its first treatment *change* F_g and follow it ℓ periods after.

Switchers
Groups whose treatment first changes at F_g.

Later changes are part of their path.

Treated group in this comparison.

Stayers (= not-yet-switched controls)
Units with the *same baseline* $D_{g,1}$ that have **not yet switched** by the comparison period ($F_{g'} > F_g - 1 + \ell$).

They may switch *later*, they just have not by the time we measure.

Control group in this comparison.

Why this is clean:

1. Controls have the same baseline treatment as switchers, so they are directly comparable
2. Controls have not switched yet, so their status-quo trend is uncontaminated
3. Comparisons go switch by switch, no cohort definition needed

What we estimate:

$$\text{Effect}_\ell = \overline{Y}_{\text{switchers}} - \hat{Y}^0_{\text{stayers}}$$

ℓ periods after the switch.

Same-Baseline Controls: Which Path Is a Valid Comparison?

The control matches your baseline level, not "untreated." Rises, falls, and returning to baseline are all valid.

Path	Baseline	Valid control	Admissible?
$0, 1, 1, 1$	0	baseline-0, not-yet-switched	✓ absorbing
$0, 2, 2, 2$	0	baseline-0, not-yet-switched	✓ intensity
$0, 2, 0$ (pulse)	0	baseline-0, not-yet-switched	✓ reverts, no cross
$2, 3, 3$	**2**	**baseline**-2, not-yet-switched	✓ same-base $\neq$ untreated
$2, 1, 1, 1$	2	baseline-2, not-yet-switched	✓ de-treatment (falls)

- $2, 3, 3$: controls are the groups *still at 2*. That is the key innovation (match on the baseline *level*, not on "untreated").
- $0, 2, 0$: the group returns to baseline, yet the effect is still the *whole path vs. staying at 0*, and dCH keeps tracking the lingering effect.
- $2, 1, 1, 1$: a switch *down* (de-treatment): controls are the groups still at 2; the effect is that of *leaving* the dose vs. staying at 2 (sign $S_g = -1$). Only *crossing* the baseline is ruled out. See the No-Crossing slide.

Reading the Control Period by Period

(a) A transitory pulse $0 \to 2 \to 0$, and you still estimate the whole path:

t	1	2	3	4
Switcher (F_g=2)	0 base	2 (ℓ=1)	0 (ℓ=2)	0 (ℓ=3)
Control	0	0 ✓	0 ✓	0 ✓

At ℓ=2, 3 the treatment is back at 0, but the effect is the *whole path vs. staying at 0*: the *lingering* effect of the pulse. The control stays baseline-0 throughout.

(b) How far can ℓ go? The control runs out:

t	1	2	3	4	5
Switcher (F_g=4)	1	1	1 base	2 (ℓ=1)	2 (ℓ=2)
Control	1	1	1	1 ✓	**2** ×

ℓ=1 is estimable (control still at 1); ℓ=2 is not (the control has switched). $\Rightarrow T_g = 4,\ \ell_{\max} = T_g - F_g + 1 = 1.$

What We Estimate: Actual vs. Status Quo

Definition $\delta_{g,\ell}$, the actual-versus-status-quo effect (p. 5)

$$\delta_{g,\ell} = \mathbb{E}\big[\, Y_{g,F_g-1+\ell} - Y_{g,F_g-1+\ell}(\underbrace{D_{g,1},\ldots,D_{g,1}}_{\text{status quo}}) \mid D \,\big]$$

For group g, ℓ periods after its last pre-switch period $F_g - 1$: the outcome under its actual dose path minus the outcome had it *kept its baseline* dose $D_{g,1}$ throughout. The part of the group's evolution that is due to the change in dose.

"Versus status quo" = compared to "keep doing what you were doing," not to "never treated."

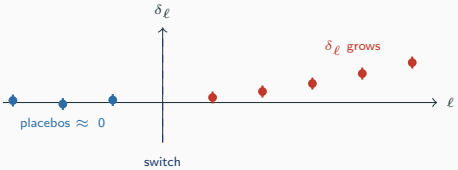
The switchers' / planner's view

The estimand is **conditional on the realized design** D (like a LATE).

The effects you estimate are those of the groups that actually switched.

If no group with a given baseline ever switches, there is nothing to compare for it: the parameter is defined by the groups that moved.

The Event Study: δ_ℓ



One number per horizon $\ell = 1, \ldots, L$. Groups are aligned by time since their own switch, so the same ℓ mixes calendar years. N_ℓ falls with ℓ: g counts only while $F_g - 1 + \ell \leq T_g$, so late switchers and groups whose same-baseline controls have all switched drop out. Each δ_ℓ averages a different set of groups.

Definition δ_ℓ, the event-study effect at horizon ℓ (p. 6, eq. 2)

$S_g = +1$ if the first change raises the dose, -1 if it lowers it. N_ℓ: switchers that still have a not-yet-switched same-baseline control, $F_g - 1 + \ell \leq T_g$.

$$\delta_\ell = \frac{1}{N_\ell} \sum_{g:\, F_g - 1 + \ell \leq T_g} S_g \, \delta_{g,\ell}$$

The average of $\delta_{g,\ell}$ over the groups still observed at horizon ℓ, switchers whose dose fell entering with the sign flipped: the effect of having received a weakly higher dose for ℓ periods.

Example, $\ell = 3$: A switched in 1996 and contributes its 1998 effect, B switched in 1999 and contributes its 2001 effect; a group that switched in 2004 in a panel ending in 2005 is not in N_3.

Placebos and Path-Specific Effects

Definition δ^{pl}_{ℓ}, the placebo at horizon ℓ (p. 11)

The same construction as δ_ℓ on the ℓ periods *before* the switch: the change $Y_{g,F_g-1-\ell} - Y_{g,F_g-1}$ of the switchers against the same change of their controls. Under no anticipation and parallel trends its expectation is zero. It tests parallel trends over as many periods as δ_ℓ needs (ℓ), with the same controls, but before the switch, not after it.

Definition path-specific δ_ℓ (pp. 6–7, fn. 10)

The average in δ_ℓ restricted to the groups that share the same dose path $(D_{g,1}, D_{g,F_g}, \ldots, D_{g,F_g-1+\ell})$: one event study per path. Useful when a few paths cover most switchers; each path then has fewer groups.

Three levels of the same object. $\delta_{g,\ell}$: one group, one horizon. Path-specific δ_ℓ: the groups that did the same thing. δ_ℓ: all the groups observed at that horizon.

The $\text{DID}_{g,\ell}$ Estimator

$\delta_{g,\ell}$ is the estimand, never observed. $\text{DID}_{g,\ell}$ is the estimator, built from changes seen in the data; Lemma 1 links the two.

Definition $\text{DID}_{g,\ell}$, the estimator of $\delta_{g,\ell}$ (p. 5)

$$\text{DID}_{g,\ell} = \underbrace{\left[Y_{g,F_g-1+\ell} - Y_{g,F_g-1}\right]}_{\text{switcher } g\text{'s change}} - \underbrace{\frac{1}{N^g_{F_g-1+\ell}} \sum_{\substack{g': D_{g',1}=D_{g,1} \\ F_{g'}>F_g-1+\ell}} \left[Y_{g',F_g-1+\ell} - Y_{g',F_g-1}\right]}_{\text{same-baseline, not-yet-switched controls}}$$

- Term 1: g's outcome change from baseline $F_g - 1$ to horizon $F_g - 1 + \ell$.
- Term 2: the average change over the *same window* among controls with the **same baseline** $D_{g,1}$ that have **not switched** by $F_g - 1 + \ell$.
- N^g_t = number of such controls.

The key innovation
Controls match on the *baseline level*, not on "untreated." Groups all at $D{=}2$ in period 1 form their own comparison set.

Unbiasedness (Lemma 1)
Under no-anticipation + parallel trends on the status-quo outcome, $\mathbb{E}[\text{DID}_{g,\ell} \mid D] = \delta_{g,\ell}$.

Key Assumptions

(a) Parallel trends on the status-quo outcome
Among groups with the same baseline $D_{g,1}$, the *status-quo* outcome ("keep $D_{g,1}$") evolves in parallel in every period (A2, p. 5).

Weaker than parallel trends across *different* baselines (A3, p. 7), which rules out lagged and time-varying effects; A2 leaves them free.

(b) No anticipation
No outcome depends on future treatments: $Y_{g,t}(d_1, \ldots, d_T) = Y_{g,t}(d_1, \ldots, d_t)$ (A1, p. 5).

Enables pre-treatment placebo tests.

(c) Lagged effects are allowed
$Y_{g,t}$ may depend on the *full treatment history*, so lagged exposure, learning, habit formation.

This is the key advantage for non-absorbing designs, where past treatment plausibly still shapes today's outcome.

Design requirement
Some baseline level must contain groups that switch at *different* dates (design restriction 1, p. 4); baseline levels without such a pair contribute nothing. If all units switch simultaneously, identification fails.

The No-Crossing Condition

Design restriction 2 (p. 6): each group's treatment must stay on *one side* of its baseline: either $D_{g,t} \geq D_{g,1}$ for all t (only rises), or $D_{g,t} \leq D_{g,1}$ for all t (only falls). It must never cross back over $D_{g,1}$.

Why crossing breaks things

A group with path $2, 3, 1$ (baseline 2) goes *above* (3) then *below* (1) its baseline. Its $\delta_{g,\ell}$ can then be written with negative weights and may flip sign even if every true effect is monotone, violating no-sign-reversal.

Intuition: the estimator nets effects against the baseline. If a group wanders to both sides, the above- and below-baseline effects get tangled with opposing signs.

When it holds automatically

- Binary treatment
- A single treatment change
- $D_{g,1} = 0$ for all groups

If violated

The cells where g has been both strictly above and strictly below its baseline are dropped (fn. 8, p. 6).

Switchers-In vs. Switchers-Out

Switchers come in two kinds, by the direction of the first change.

The sign of the switch

$$S_g = \mathbf{1}\{D_{g,F_g} > D_{g,1}\} - \mathbf{1}\{D_{g,F_g} < D_{g,1}\}$$

$S_g = +1$: **switcher-in** (dose rises above baseline).
$S_g = -1$: **switcher-out** (dose falls below baseline).

Definition DID_ℓ, the estimator of δ_ℓ (p. 6; companion eq. 5)

Two event studies, $\mathrm{DID}_{+,\ell}$ over the N_ℓ^1 switchers-in and $\mathrm{DID}_{-,\ell}$ over the N_ℓ^0 switchers-out (built on $-\mathrm{DID}_{g,\ell}$), then aggregated:

$$\mathrm{DID}_\ell = \frac{N_\ell^1}{N_\ell^1+N_\ell^0}\mathrm{DID}_{+,\ell} + \frac{N_\ell^0}{N_\ell^1+N_\ell^0}\mathrm{DID}_{-,\ell}.$$

Why flip the sign for switchers-out?

After the flip, *every* term reads as "the effect of being exposed to a **weakly larger dose** for ℓ periods." Ins and outs are placed on the same footing, so DID_ℓ aggregates comparable quantities.

Pooling ins and outs

DID_ℓ averages the effect of adding dose with the sign-flipped effect of removing it. If they may differ (a plant closing hurts more than its opening lifts), estimate them **separately**: `switchers(in)`, `switchers(out)`.

The Dose Accumulated: $\delta^D_{g,\ell}$ and δ^D_ℓ

With a non-binary treatment, δ_ℓ averages groups whose extra dose differs in size and in duration (p. 6): a $0 \to 1$ switch and a $0 \to 4$ switch enter as one switcher each. To compare them the effect is divided by the amount of treatment received (p. 7).

Definition $\delta^D_{g,\ell}$, the dose accumulated up to horizon ℓ (pp. 7–8)

$$\delta^D_{g,\ell} = \sum_{k=0}^{\ell-1} \left(D_{g,F_g+k} - D_{g,1}\right), \qquad \delta^D_\ell = \frac{1}{N_\ell} \sum_{g:\, F_g - 1 + \ell \le T_g} \left|\delta^D_{g,\ell}\right|$$

Every dose-period received from the switch to $F_g - 1 + \ell$, summed; and its average over the switchers observed at ℓ. A quantity of treatment, computed from D, with no estimation error.

The unit: a dose-period
One unit of D in force during one period. A binary policy: one period under the policy. A count of laws: a law-year. A share in points: a point-year.

A group that goes from 0 to 2 and stays: in force $2, 2, 2$; accumulated $2, 4, 6$. If it moves to 3 in its third period: $2, 4, 7$. Binary absorbing: accumulated $= \ell$.

The Normalized Effect: $\delta^n_{g,\ell}$ and δ^n_ℓ

Definition $\delta^n_{g,\ell}$ and δ^n_ℓ, the effect per dose-period (pp. 7–8, eq. 4)

$$\delta^n_{g,\ell} = \frac{\delta_{g,\ell}}{\delta^D_{g,\ell}}, \qquad \delta^n_\ell = \frac{\delta_\ell}{\delta^D_\ell}$$

The effect divided by the dose accumulated up to ℓ: what one dose-period has added by horizon ℓ, in outcome units per dose-period, comparable across groups whatever their dose.

Lemma 2 (p. 8): what it averages

$$\delta^n_{g,\ell} = \sum_{k=0}^{\ell-1} w_{g,\ell,k}\, s_{g,\ell,k}$$

a weighted average of the slopes (effect per unit of dose) with respect to the current dose ($k = 0$) and its $\ell - 1$ lags, with weights $w_{g,\ell,k}$ proportional to the extra dose at each lag, summing to one and non-negative under design restriction 2. One change kept in force: $1/\ell$ each.

The lags cannot be separated unless lag effects are additive, linear and constant over time (fn. 15); the weights $w_{\ell,k}$ can be reported (p. 9). A flat profile $\ell \mapsto \delta^n_\ell$ says current and lagged doses have similar effects (p. 17); with one change kept in force, a profile falling as $1/\ell$ says only the dose in force matters.

In Words (1): From the Raw Effect to the Normalized Effect

Raw. $\delta_{g,\ell}$ is the group's actual outcome against its status quo, ℓ periods after the switch; δ_ℓ is the same thing averaged over the switchers observed at ℓ. With a binary absorbing treatment nothing more is needed: every switcher took the same dose, one unit.

Why a non-binary treatment needs more. Switchers differ in how much dose they took, and the dose stays in force period after period, so the effect measured at ℓ reflects both how much and for how long. δ_ℓ averages a $0 \to 1$ county and a $0 \to 4$ county as one switcher each.

Accumulated dose. $\delta^D_{g,\ell}$ counts the dose-periods the group has received since its switch: $0 \to 2$ and staying gives $2, 4, 6$. δ^D_ℓ is the *average* of those counts over the switchers observed at ℓ, an average like δ_ℓ, not a sum. A second county that moves $0 \to 1$ one period later has $1, 2$ at its own $\ell = 1, 2$ and is not observed at $\ell = 3$: δ^D_ℓ is $(2+1)/2 = 1.5$, then $(4+2)/2 = 3$, then 6.

Normalized. $\delta^n_\ell = \delta_\ell / \delta^D_\ell$, the raw effect divided by the accumulated dose: the effect per dose-period. Numerator and denominator average over the same groups, so it also equals the sum of the groups' sign-adjusted effects $S_g \delta_{g,\ell}$ over the sum of their absolute dose-periods $|\delta^D_{g,\ell}|$, and it is a $|\delta^D_{g,\ell}|$-weighted average of each group's own $\delta_{g,\ell}/\delta^D_{g,\ell}$ (eq. 4).

The Average Total Effect per Unit of Treatment, δ

Definition δ, the average total effect per unit of treatment (p. 9)

$$\delta = \frac{\sum_g \sum_{\ell=1}^{T_g - F_g + 1} \delta_{g,\ell}}{\sum_g \sum_{\ell=1}^{T_g - F_g + 1} \left(D_{g,F_g - 1 + \ell} - D_{g,1} \right)}$$

Every actual-versus-status-quo effect, over every horizon a switcher is observed, divided by every dose-period in force over those horizons: the total effect of one dose-period, the period it arrives and every later horizon in the window. No subscript, as in the paper: one number over all groups and horizons.

With no discounting and a cost linear in the dose, the switchers' actual treatments are beneficial iff $\delta > c$, the average cost of one dose-period (p. 9). Lemma 3 (p. 10): under design restrictions 1 and 3 (no group ever goes below its baseline, p. 9), $\delta = \sum_\ell w_\ell \, \delta_\ell$ with non-negative weights, so if every δ_ℓ has the same sign, δ has it too.

Definition the average number of periods over which a dose's effect is cumulated (p. 10)

$$\frac{\sum_g \sum_{k=0}^{T_g - F_g} \left(D_{g,F_g + k} - D_{g,1} \right) \left(T_g - F_g - k + 1 \right)}{\sum_g \sum_{k=0}^{T_g - F_g} \left(D_{g,F_g + k} - D_{g,1} \right)}$$

Each dose-period weighted by the number of horizons it is counted in. Dividing δ by it gives the average effect of one dose for one period.

Binary staggered with a never-treated group (p. 9): δ is the average, over treated (g, t) cells, of the effect of having been treated for $t - F_g + 1$ periods, the ATT generalized to dynamic effects; and $\delta_\ell^n = \delta_\ell / \ell$.

In Words (2): The Average Total Effect per Unit of Treatment

A sum, not a division horizon by horizon. The numerator adds every raw effect $\delta_{g,\ell}$ of every switcher at every horizon it is observed. The denominator adds the extra dose *in force* at each of those group-horizons: the total dose-periods received over the window. The county at $0 \to 2$ for three periods contributes $2 + 2 + 2 = 6$ dose-periods; the county at $0 \to 1$ for two periods contributes 2; eight in all.

In force, not accumulated. $\delta^{D}_{g,\ell}$ already contains the earlier periods; summing it over ℓ would count the first dose-period three times. δ sums the flow of each period, which equals the group's accumulated dose at its last observed horizon.

The same fraction per cell. Divide numerator and denominator by the number of group-horizon cells: δ is the average raw effect per cell divided by the average dose in force per cell. Binary absorbing treatment: the dose in force is 1, and δ is the average of the δ_ℓ weighted by the switchers at each horizon. With a dose, the same average divided by the mean dose in force.

Against the normalized effect. Both divide by dose. δ^{n}_{ℓ} divides one horizon's effect by the dose accumulated up to it: what a dose-period adds per period. δ divides all effects by all dose-periods, and since a dose-period keeps being counted in later horizons, it gives what a dose-period adds in total over the window. $\delta = M\times$ the average of the δ^{n}_{ℓ} weighted by the dose accumulated at ℓ, with M the average number of periods over which a dose's effect is cumulated (p. 10; derived from Lemma 3). δ/M equals every δ^{n}_{ℓ} exactly when the profile is flat.

dCH vs. C&S: When to Use Which

Feature	C&S	dCH
Binary treatment	Yes	Yes
Non-binary / intensity	No	**Yes**
Absorbing treatment	Yes	Yes
Non-absorbing	No	**Yes**
Doubly robust	**Yes**	No
Covariate-adj. PT	**Yes**	Linear only
Default SEs (Stata)	Analytic	Analytic

What each one requires
C&S requires a binary, absorbing treatment; when all units start untreated, dCH's assumption 2 is similar to its parallel trends (p. 5). dCH allows any design satisfying restriction 1 (p. 4).

When every unit starts untreated, DID_ℓ equals the C&S event study at $\ell - 1$ computed on the binarized and staggerized treatment (dCH 2024 *REStat*, p. 7).

Decision rule:

- **Binary and absorbing?**
  Both valid. Use dCH as a robustness check for C&S. If they agree, C&S is preferred (doubly robust).
- **Non-binary (intensity/count)?**
  dCH only
- **Non-absorbing (reversible)?**
  dCH only, allowing full treatment history to affect Y
- **Covariates important?**
  C&S has the stronger doubly-robust estimator

Extensions (1): Relaxing Parallel Trends

The baseline parallel-trends assumption can be weakened in several ways. Rather than list them abstractly, we introduce each *in the case that needs it*, where a concrete failure makes the fix easy to motivate:

- **Covariate-adjusted** trends, `controls()`: Deryugina (Case 2).
- **Trends within sets of groups**, `trends_nonparam()`: Hollingsworth (Case 5).
- **Group-specific linear** trends, `trends_lin`.
- A smooth **polynomial** trend for a continuous baseline: next slide.

Extensions (2): Continuous Treatment

When the period-one treatment $D_{g,1}$ is continuously distributed, no two groups share a baseline, so same-baseline matching has nothing to work with.

Model the trend in the baseline

Instead of matching on an identical $D_{g,1}$, assume the status-quo trend is a smooth (polynomial) function of the baseline:

$$\mathbb{E}\big[\Delta Y^{\mathsf{sq}}_{g,t} \mid D\big] = \sum_{k=0}^{K} \gamma_{k,t}\, D^{k}_{g,1},$$

with the order K chosen by the researcher. This trend is netted out, and the estimator proceeds as before.

What it identifies

By default, the **level** effect of having changed the dose: the numerator uses a binarised switch (companion, pp. 22–23). Dividing by the actual change in D, in the normalized effect and in δ, gives an effect **per unit of dose**: a scaled level, not a derivative at a point.

Inference

Variance is conservative when K is right, possibly liberal when too high, so inference relies on the bootstrap.

A topic of its own, developed in later work (de Chaisemartin, D'Haultfœuille, Pasquier & Vazquez-Bare 2022; Callaway, Goodman-Bacon & Sant'Anna, forthcoming *AER*). We use the dCH option for it in Case 3, and take up the question properly in the continuous-treatment block.

Running did_multiplegt_dyn: One Command, Step by Step

```
* ssc install did_multiplegt_dyn    // once
did_multiplegt_dyn Y G T D, ///
    effects(8) placebo(3) cluster(state_n)
```

The four arguments, left to right:

- `Y`: the outcome.
- `G`: the group, the unit whose outcome you follow, with one treatment value per group and period: the *county* in Favara, though the law is set by state; the state goes to `cluster()`.
- `T`: the time variable.
- `D`: the treatment.

The three base options:

- `effects(8)`: the switch period and the 7 *after*.
- `placebo(3)`: check 3 periods *before* (must be $\leq$ `effects`).
- `cluster(state_n)`: SEs by state with $g =$ county; the default is g, and only a coarser level is allowed, never finer.

Paper	Output	Option
δ_ℓ, event-study effect	`Effect_`ℓ, with N (observations, fn. 4) and Switchers (groups)	default
placebo effect	`Placebo_`ℓ	`placebo(#)`
$\delta^n_{\ell'}$, per dose-period (Lemma 2)	`Effect_`ℓ	`normalized`
lag weights $w_{\ell,k}$	table $k \times \ell$	`normalized normalized_weights`
δ, total per dose-period (p. 9)	`Av_tot_eff`; also `Switch x Periods` and the average number of periods over which a dose's effect is cumulated (p. 10)	default; not with `trends_lin`

How to Read Each Number, in One Sentence

Raw `Effect_`ℓ. In the ℓ-th period counting the period of their first change, so $\ell - 1$ periods after it, switchers' outcome is on average [Effect$_\ell$] [units of Y] higher than had they kept their initial dose ("lower" if their dose fell, $S_g = -1$, p. 6).
En el periodo ℓ contando el de su primer cambio, es decir $\ell - 1$ periodos después, el outcome de los switchers es en media [Effect$_\ell$] [unidades] mayor que con su dosis inicial ("menor" si su dosis bajó).

`Effect_`ℓ with `normalized`. At horizon ℓ, each dose-period accumulated since the switch accounts for [Effect$_\ell$] [units of Y], an average over the current dose and its lags.
En el horizonte ℓ, cada dosis-periodo acumulada desde el cambio explica [Effect$_\ell$] [unidades], media de la dosis actual y las rezagadas.

`Av_tot_eff`. One unit of dose in force for one period changes the outcome by [Av_tot_eff] [units of Y] in total over the `effects()` window.
Una unidad de dosis en vigor durante un periodo cambia el outcome en [Av_tot_eff] [unidades] en total a lo largo de la ventana.

`Placebo_`ℓ. In the ℓ periods before the change, switchers and their controls moved alike (joint $p = \ldots$), or apart by [Placebo$_\ell$].
En los ℓ periodos anteriores al cambio, switchers y controles se movían igual (p conjunto $= \ldots$).

Units. Those of Y: a log outcome reads in percent ($0.011 \approx 1.1\,\%$); a dose in its own unit (law-years, newspaper-elections, a unit of eligibility, 100 points, for a year). Binary absorbing: `Effect_`ℓ is the effect of ℓ periods under treatment, `normalized` divides it by ℓ, `Av_tot_eff` is the switcher-weighted mean of the raw effects. Binary pulse, one period in force: `Av_tot_eff` sums the effects, each weighted by the share of switchers left at that horizon.

Applications

Five Cases

One engine, matching on the baseline and comparing switchers to not-yet-switchers, spans a whole family of designs.

	Paper	Treatment
1	Favara & Imbs (2015)	Count 0–4 (restrictions lifted)
2	Deryugina (2017)	Binary non-absorbing
3	East et al. (2023)	Continuous (Medicaid %)
4	Gentzkow et al. (2011)	Count (newspapers)
5	Hollingsworth et al. (2022)	Sequential binary (mm $\rightarrow$ rm)

Full code: `Stata/Block2_dCH_practice.do`

Case 1 — Favara & Imbs (2015): Context & Data

Research question
Does credit supply (bank branching deregulation) raise mortgage lending?

Paper
Favara & Imbs (2015), *AER* 105(3): 958–992.

Data (replication package)
1,048 US counties $\times$ **12 years** (1994–2005); 12,538 obs; clustered by state (50).

Outcome `Dl_vloans_b` (log change in mortgage volume); treatment `inter_bra` (interstate-branching restrictions lifted; reversed Rice & Strahan index, 0–4).

Design (read off the data)

- **Count** treatment $\in \{0, 1, 2, 3, 4\}$ (Rice & Strahan index).
- **99.9%** of counties start at 0 (first observed year).
- Mostly accumulating: 1,056 increases vs. 46 decreases.
- **917** counties change their index; **905** enter the estimation at $\ell = 1$.

Non-binary, so TWFE makes an intermediate-dose county both treated and control; dCH with `normalized` gives the effect per additional restriction lifted.

Case 1 — The Output, Block by Block

```
Estimation of treatment effects: Event-study effects
            |  Estimate        SE      LB CI      UB CI       N
  Effect_1  |  .0434814  .0354981  -.0260935   .1130563    3810    (1)
  ...
  Effect_8  |   .185827  .0794133   .0301799   .3414741    1248
            | Switchers
  Effect_1  |       905     ...     Effect_8  |     773            (2)
Test of joint nullity of the effects : p-value = .0635             (3)

Average cumulative (total) effect per treatment unit
Av_tot_eff  |  .0478284  .0246733  -.0005305   .0961873   10598    (4)
            | Switch x Periods
Av_tot_eff  |      6921                                           (5)
Average number of time periods over which a treatment's
  effect is accumulated = 4.3392959                                 (6)

Testing the parallel trends and no anticipation assumptions
 Placebo_1  |  .0548088  .0511118  -.0453685   .154986    2777    (7)
  ...
Test of joint nullity of the placebos : p-value = .4068
```

1. Raw Effect$_\ell$, in the outcome's units (log points of loan growth); with `normalized`, per law-year. *N*: observations used.
2. Switchers observed ℓ periods after switching; the count falls once no same-baseline control is left.
3. Joint test of the eight effects.
4. `Av_tot_eff`: total effect per law-year over the window. Identical with or without `normalized`.
5. Switcher-periods used: $\sum_\ell \text{Switchers}_\ell = 6{,}921$.
6. Average number of horizons a law-year's effect is counted in, from the dose paths, each county's last year with a same-baseline control and `effects(8)`; it links the per-year and total effects.
7. Placebos, the mirror of the effects before the switch, and their joint test.

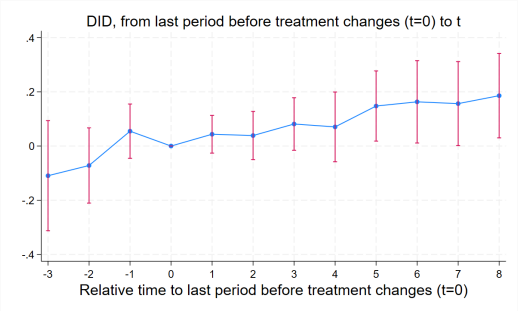
Case 1 — Raw Effects

```
did_multiplegt_dyn      ///
  Dl_vloans_b           ///
  county year inter_bra ,///
  effects(8) placebo(3) ///
  cluster(state_n)
```

Raw effects, in log points of loan growth:

- 0.043 at $\ell = 1$, 0.186 at $\ell = 8$; Effects$_{5-8}$ individually significant
- Joint $p = 0.064$; placebos $p = 0.407$ ✓

The profile rises because law-years accumulate; a $0 \to 1$ and a $0 \to 4$ county count as one switcher each.
No-Crossing holds: no county falls below its initial index.



Reading

In the eighth year counting the year of the first step, seven years after it, annual loan growth is 18.6 log points above what it would have been had the county kept its initial index, for a county carrying 18.4 accumulated law-years on average.
En el octavo año contando el del primer paso, siete años después, el crecimiento anual del crédito es 18.6 puntos logarítmicos mayor de lo que habría sido de haber mantenido el índice de partida, en un condado que acumula 18.4 ley-años de media.

Case 1 — From Raw to Normalized: Eight Effects Divided by the Dose

Two runs of the same command, without and with `normalized`. The command does not print δ_ℓ^D; dividing one run by the other recovers it, and consecutive differences approximate the extra laws in force at each horizon.

ℓ	Raw DID$_\ell$	Normalized DID$_\ell^n$	Ratio $= \delta_\ell^D$, law-years	Difference	Switchers
1	0.0435	0.0197	2.2	2.2	905
2	0.0387	0.0088	4.4	2.2	905
3	0.0813	0.0123	6.6	2.2	905
4	0.0706	0.0080	8.8	2.2	905
5	0.1476	0.0134	11.0	2.2	905
6	0.1631	0.0121	13.5	2.4	850
7	0.1565	0.0098	16.0	2.5	773
8	0.1858	0.0101	18.4	2.5	773

The switcher mean rose by 2.2 laws at its first change and kept them, so the accumulated column grows by 2.2 a year while the switchers are the same 905. The difference rises afterwards for two reasons, checked in the data: 101 counties received further steps after their first change (the extra laws in force go $2.20, 2.19, 2.19, 2.20, 2.24, 2.28, 2.33, 2.47$), and from $\ell = 6$ the late switchers leave while the early ones had a slightly larger first change (2.25).

Case 1 — Normalized Effects: Per Law-Year

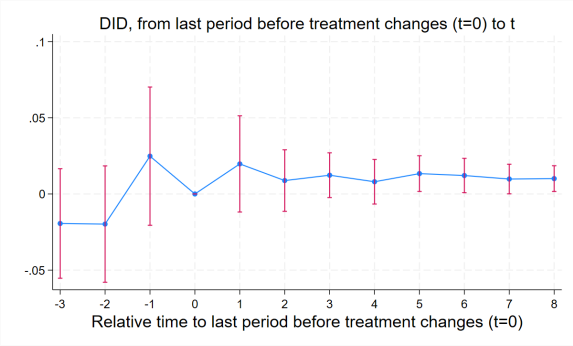
```
... effects(8) placebo(3) cluster(state_n)
normalized normalized_weights
```

Reading

At $\ell = 8$, each law-year received so far accounts for one log point of annual loan growth (0.010 in log units): one step adds about one point per year in force, flat across horizons.

En el octavo año, cada ley-año recibido hasta entonces explica un punto logarítmico del crecimiento anual del crédito (0.010 en unidades log): un paso añade un punto por cada año en vigor, igual en todos los horizontes.

Effects$_{5-8}$ significant at 5%; joint $p = 0.064$; placebos $p = 0.41$. The raw profile quadruples between $\ell = 1$ and 8 while the exposure grows eightfold. Switchers fall from $\ell = 6$ because the panel ends in 2005 (`same_switchers` holds the sample fixed).



Case 1 — `Av_tot_eff`, Step by Step

Numerator: the raw effects, added over the window
$0.043 + 0.039 + 0.081 + 0.071 + 0.148 + 0.163 + 0.157 + 0.186 \approx 0.887$ (exact sum 0.8871), unweighted; the command weights each horizon by its switchers.

Denominator: the extra laws in force, added over the window
$2.2 + 2.2 + 2.2 + 2.2 + 2.2 + 2.4 + 2.5 + 2.5 = 18.4$ law-years from the table.

$$\hat{\delta} \approx \frac{0.887}{18.4} = 0.048 \quad \text{(command, switcher-weighted: 0.0478)}$$

One law-year raises loan growth by 4.8 log points in total over the eight horizons.

The output line "average number of time periods over which a treatment's effect is accumulated $= 4.34$"
The paper's p. 10 quantity, computed from the dose paths, the last year each county has a same-baseline control and `effects(8)`: a law-year arriving at horizon 1 of a county observed all eight horizons is counted 8 times, one arriving at horizon 8 once, and counties switching in 2000 or 2001 are observed fewer horizons before 2005. The dose-weighted average is 4.34.

$$\frac{0.048}{4.34} = 0.011 \approx \text{the flat normalized level}$$

Per law-year and per year, 0.011; counted over 4.34 horizons on average, 0.048. A shorter `effects()` lowers both the 4.34 and $\hat{\delta}$.

Case 1 — `Av_tot_eff`: What It Says and What to Report

Reading

One deregulation step in force for one year raises annual loan growth by 4.8 log points in total over the eight-year window, about 1.1 per year over the 4.3 years it is counted.
Un paso de desregulación en vigor un año eleva el crecimiento anual del crédito 4.8 puntos logarítmicos en total a lo largo de la ventana de ocho años, 1.1 por año durante los 4.3 años en que se cuenta.

Which number is the headline
SE 0.025, CI $[-0.001,\ 0.096]$, just including zero while four horizons are individually significant; identical with or without `normalized`. The normalized profile is flat, so $\hat{\delta}$ summarizes it and the event study is the evidence that the design holds: placebos at $p = 0.41$, effects growing with the law-years and stable per law-year.

For your own design
Raw: outcome units at horizon ℓ, actual path against status quo. Normalized: per dose-period accumulated up to ℓ; recover the accumulated dose by dividing the two runs. `Av_tot_eff`: per dose-period, cumulated over your `effects()` window; set it against the cost of one dose-period, and report the window with it. Binary absorbing: normalized $=$ raw$/\ell$ and `Av_tot_eff` $=$ the mean of the raw effects weighted by switchers per horizon.

Case 1 — `normalized_weights`: Weights on Treatment Lags

A normalized Effect$_\ell$ blends the effect of the *current* treatment (k=0) and its lags (k=1, 2, ...). The lags cannot be separated, but the *weight* on each can be reported:

k	ℓ=1	ℓ=2	ℓ=4	ℓ=8
0	**1.00**	0.50	0.25	0.13
1	—	0.50	0.25	0.13
2	—	—	0.25	0.13
3	—	—	0.25	0.12
⋮				⋮
7	—	—	—	0.12
Tot.	1.00	1.00	1.00	1.00

How to read it

- Effect$_1$: all weight on k=0: only the current dose, no lags yet.
- Effect$_2$: half on the current dose, half on the 1-period lag.
- Effect$_8$: $\approx$ 0.12–0.13 on *each* of k=0, ..., 7 , nearly uniform.

Each weight is the share of the law-years received so far that comes from that lag: $\approx 1/8$ each at $\ell = 8$ because a law, once passed, stays. So Effect$_8$ is the average contribution of a law-year over eight years of exposure, not the effect of a law eight years after it passed.

Applied conclusion: Favara's pair (raw rising with the law-years, normalized flat) is the persistent case of the worked example. Whether one step fades or builds needs `by_path` or `normalized effects_equal`.

Case 4 — Gentzkow et al. (2011): Context & Data

Research question
Do local daily newspapers affect voter turnout?

Paper
Gentzkow, Shapiro & Sinkinson (2011), *AER* 101(7): 2980–3018.

Data (replication package)
Unbalanced panel of **1,195 US counties** $\times$ **16 presidential-election years** (1868–1928); 16,872 obs.

Outcome `prestout` (turnout rate); treatment `numdailies` (number of daily papers).

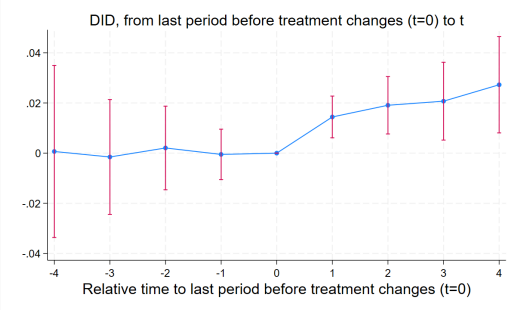
Design (read off the data)

- Baseline treatment is **multivalued** (0 to 33; 75.6% of counties start at 0).
- Counties **gain and lose** papers (Δ both $+$ and $-$).
- **81%** of counties change their count **two or more times**.

A multivalued, non-absorbing treatment: no single first-treatment date, so identify from each change and normalize by switch size.

Case 4 — Effects Significant, Pre-Trends Flat

```
did_multiplegt_dyn      ///
  prestout              ///
  cnty90 year           ///
  numdailies,           ///
  effects(4) placebo(4) ///
  cluster(cnty90)       ///
  effects_equal("all")
```



Results:

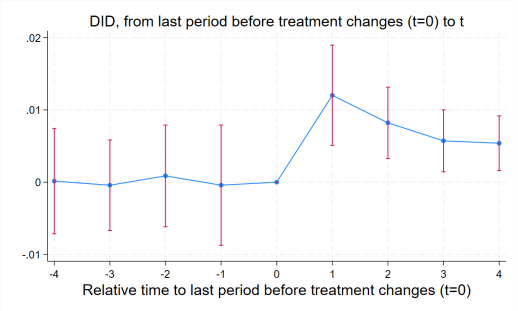
- Effects: 0.014, 0.019, 0.021, 0.027; joint $p = 0.007$
- effects_equal $p = 0.415$ (stable over time)
- Placebos: −0.001, 0.002, −0.002, 0.001; **placebo** $p = 0.992$ ✓✓✓

Reading

In the election in which the number of dailies changes, by 1.2 on average, turnout is 1.4 points higher than it would have been with the previous count, on a 65 % base; 2.7 points by the fourth election, three elections later.
En la elección en que cambia el número de diarios (1.2 de media), la participación es 1.4 puntos mayor de lo que habría sido con el número anterior, sobre una base del 65 %; 2.7 puntos en la cuarta elección, tres después.

Case 4 — Normalized: Effect per Newspaper-Election

```
did_multiplegt_dyn      ///
  prestout cnty90       ///
  year numdailies,      ///
  effects(4) placebo(4) ///
  cluster(cnty90)       ///
  normalized            ///
  normalized_weights    ///
  effects_equal("all")
```



Per newspaper-election: effects 0.012, 0.008, 0.006, 0.005; placebo $p = 0.992$; `Av_tot_eff` $= 0.016$.

The decline is not significant ($p = 0.165$). It suggests lagged papers matter less than current ones (companion, pp. 39–40); a direct test that the first lag has no effect gives $p = 0.83$ (p. 40).

Reading

Per newspaper-election, turnout is 1.2 points higher in the election of the change and 0.5 by the fourth, three elections later; one newspaper in circulation for one election raises turnout 1.6 points in total over the four-election window.

Por periódico-elección, la participación sube 1.2 puntos en la elección del cambio y 0.5 en la cuarta, tres después; un periódico en circulación durante una elección sube la participación 1.6 puntos en total en la ventana de cuatro elecciones.

Case 4 — Treatment Paths and Composition

Which paths identify the effect? (`design`)

- The estimate pools *many* distinct treatment paths.
- Most common: $0 \to (1,1,1,1)$, 15% of the 933 switchers observed four elections; next, enter-then-exit, 14%.
- The listed paths cover 80% of them.

Transparency about *what variation* drives the estimate, like `normalized_weights` earlier.

Composition (`same_switchers`)

Holding the sample fixed at the **907 switchers** present at every horizon: effects $0.013 \to 0.029$ (joint $p = 0.019$), placebo $p = 0.93$.

Close to the baseline, so the growing event study is **not** a composition artefact.

Running dCH — The Practical Answers

At what level do I cluster?

At g or coarser, never finer: with $g =$ county and a state policy, `cluster(state)`; with $g =$ state, a state weighs its number of counties.

How many effects and placebos?

`placebo()` must not exceed `effects()`. The horizon is capped by the data at $\ell_{\max} = T_g - F_g + 1$, and the switcher count falls as ℓ grows. Read that column before you read the coefficients.

`normalized` or not?

If the dose varies in size, yes: without it a $0 \to 1$ and a $0 \to 4$ switch are averaged as the same event. The normalized effect averages over doses *and* lags, with the lag weights printed by `normalized_weights`. When a few paths dominate, `by_path()` gives each path's effect against the status quo with no normalization; the price is switchers per path.

Which number goes in the report?
Raw Effect$_\ell$: outcome units at horizon ℓ, actual path against status quo. Normalized: per dose-period accumulated up to ℓ. `Av_tot_eff`: per dose-period, cumulated over the `effects()` window, the cost-benefit number (p. 9). The event study is the *evidence* that the design holds. Course rule: the headline is `Av_tot_eff` when the normalized profile is stable, the per-horizon effects when it is not.

Two things that are not errors
Wide confidence intervals when few units switch, and effects that stop at a short horizon because the panel ends. Both are the estimator reporting the precision your data actually support.

Placebos that fail, sequential treatments and continuous doses come next.

dCH So Far, in Four Lines

1. **Switchers and stayers.** Each effect compares groups whose treatment changes at F_g with groups that held the *same* period-one treatment and had not yet moved.
2. **Effect$_\ell$ and Placebo$_\ell$.** The event study reads forward from the switch; the placebos read backward and are zero under parallel trends and no anticipation.
3. **Normalization.** With a non-binary treatment, `normalized` turns the effect into an effect *per dose-period accumulated up to* ℓ, and `normalized_weights` shows how much of it comes from the current treatment and how much from its lags.
4. **Two count cases.** Favara & Imbs and Gentzkow et al.: the same command, clean placebos in both.

Next: designs where the placebos *fail*, treatments that arrive in sequence, and treatments that are a dose rather than a switch.

Case 2 — Deryugina (2017): Context & Data

Research question
How do hurricanes affect non-disaster government transfers to counties?

Paper
Deryugina (2017), *AEJ: Policy* 9(3): 168–198.

Data (replication package)
Unbalanced panel of **1,183 US counties** × **44 years** (1969–2012); 49,698 obs.

Outcome `log_curr_trans_ind_gov_pc` (log non-disaster transfers p.c.); treatment `hurricane` (0/1).

Design (read off the data)

- Binary, **non-absorbing**: each county's total hurricane count is **0 or 1**: hit in at most one year (on, then off).
- **361** of 1,183 counties are ever hit.

The cautionary tale
Before the hurricane, transfers in hit counties were already *rising* relative to unhit counties (all six placebos negative), so the placebos fail and the effects are likely overstated. dCH *diagnoses* the violation; it does not fix it.

Case 2 — First Estimates: Large, Growing, Significant

Effect of a hurricane (no controls):

ℓ	Effect	sig.?
1	0.011	✓
2	0.013	✓
3	0.016	✓
4	0.016	✓
5	0.023	✓
6	0.027	✓

Joint $p = 8 \times 10^{-13}$; **Av_tot_eff** $= 0.106$.

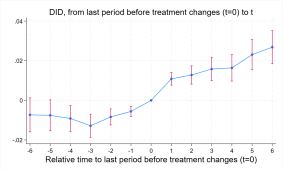
A transient shock, a persistent effect
`hurricane` is 1 for a single year, then 0. Transfers stay elevated and keep rising up to five years later; the dynamic estimator traces the carryover, which a method assuming the effect lasts only while $D = 1$ would miss.

But can we believe it?
Positive, growing, hugely significant. The question first: do the pre-treatment placebos hold?

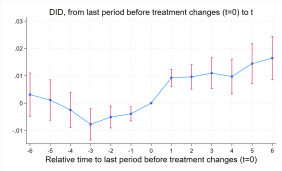
Reading

Transfers per capita are 1.1 % higher in the hurricane year than they would have been without it, and 2.7 % five years later; a hurricane raises transfers by 10.6 % of one year's transfers in total over six years.
Las transferencias per cápita son un 1.1 % mayores el año del huracán de lo que habrían sido sin él, y un 2.7 % cinco años después; un huracán eleva las transferencias un 10.6 % de las de un año en total a lo largo de seis años.

Case 2 — Pre-Trends That Fail (and Partially Improve)



Left: no controls



Right: with pre-trend controls

```
* Without controls:
did_multiplegt_dyn      ///
  log_curr_trans_ind_gov_pc ///
  county_fips year      ///
  hurricane,            ///
  effects(6) placebo(6) ///
  cluster(county_fips)

* With baseline char. x year:
did_multiplegt_dyn ..., ///
  effects(6) placebo(6) ///
  controls('controls')  ///
  cluster(county_fips)
```

Without controls:
Placebo $p = 4.3 \times 10^{-6}$ ×

With controls:
Placebo $p = 0.000035$ ×

Lesson: controls improve but do not resolve the violation. The problem is fundamental. Report it.

Case 2 — When Parallel Trends Fail: What To Do

Option 1: Add controls (char. $\times$ year)

- Allow each type of unit to follow its own linear trend
- Deryugina: p goes from 4.3×10^{-6} to 3.5×10^{-5}, still fails
- Only works when the trend difference is proportional to observables

Option 2: `trends_nonparam`

- Year effects by a time-invariant set coarser than the unit (e.g. state)
- For states or industries on different trends

Option 3: Restrict sample

- Compare only geographically or demographically similar units

Option 4: Report as limitation

- Pre-trend failure is informative : it shows *which* units drive the violation
- Reporting the PT failure is preferable to ignoring it

Key takeaway

dCH *diagnoses* the problem. It does not guarantee a fix. That requires subject-matter knowledge and, often, a better design.

Reporting the failure is preferable to leaving it undetected.

Cases 2 vs. 4 — Same Estimator, Different Pre-Trends

Same command, same country, opposite pre-trend results.

Deryugina (2017): Fails
Placebo joint $p = 4.3 \times 10^{-6}$

All placebos are negative: before the hurricane, hit counties' transfers were already *rising* relative to unhit counties, so the effect is likely overstated.

Adding controls does not resolve it.

Gentzkow et al. (2011): ✓ Passes
Placebo joint $p = 0.992$

No evidence that counties changing their newspaper count trended differently from stable counties in prior elections.

This supports parallel trends but does not prove it: the test can lack power (Roth 2022).

The pre-trend test reflects whether the design is credible; the result depends on the data, not on the researcher's choice.

When a placebo rejects, its sign suggests the bias direction, if the pre-switch trend gap continues after the switch: placebos run backward from $F_g - 1$ (p. 11), so a negative placebo suggests an upward bias.

Case 5 — Hollingsworth et al. (2022): Context & Data

Research question
How do medical (mm) and recreational (rm) marijuana laws affect marijuana use?

Paper
Hollingsworth, Wing & Bradford (2022), *J. Law and Economics* 65(3): 515–554.

Data (replication package)
51 states × **15 years** (2002–2016); 765 obs.

Outcome `ln_mj_use_365` (log past-year use); treatments `mm` and `rm` (both 0/1).

Design (read off the data)

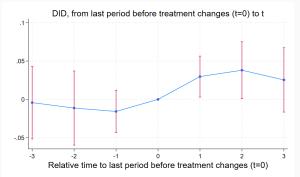
- **Two sequential binary** treatments, both **absorbing** (no reversals); `rm` only after `mm`.
- **29** states adopt mm; only **7** adopt rm.

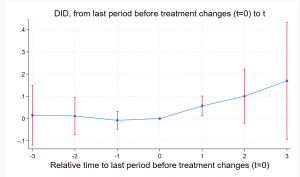
Estimate each separately: mm on states without rm; rm on states with mm, with `trends_nonparam(fg1)` (trends by mm-adoption cohort). The few switchers foreshadow wide CIs.

Case 5 — Medical and Recreational Marijuana

```
* Effect of mm (rm==0 states):
did_multiplegt_dyn      ///
  ln_mj_use_365 state   ///
  year mm if rm==0,     ///
  effects(3) placebo(3) ///
  cluster(state)

* Effect of rm (mm==1 states):
did_multiplegt_dyn      ///
  ln_mj_use_365 state   ///
  year rm if mm==1,     ///
  effects(3) placebo(3) ///
  trends_nonparam(fg1)  ///
  cluster(state)
```





Left: medical marijuana Right: recreational

mm (21 switchers): effects 0.030, 0.038, 0.025, $p = 0.155$; placebo $p = 0.711$ ✓

rm (7 switchers!): effects 0.057–0.169, $p = 0.066$; placebo $p = 0.933$ ✓

Wide CIs = few states, not zero effect.

Reading

Past-year use is 3.0 % higher in the first year of a medical law than it would have been without one; one year of the law in force raises use 3.1 % in total over the three-horizon window; 21 switchers in year one, 16 in year three.

El consumo en el último año es un 3.0 % mayor el primer año de la ley medicinal de lo que habría sido sin ley; un año de ley en vigor eleva el consumo un 3.1 % en total en la ventana de tres horizontes; 21 switchers en el año uno, 16 en el tres.

Case 5 — Why `trends_nonparam(fg1)` for the rm Estimation

Problem: mm=1 states differ in years of mm exposure, so a change in ln_mj_use around rm adoption can come from one more year of mm.

The identification concern:

- If the mm effect varies with exposure, states with different mm dates diverge even without rm
- fg1 compares rm adopters only with non-adopters that passed mm the same year
- Assumes one more year of mm has the same effect in every state

`trends_nonparam(fg1)`
fg1 = year of first mm adoption.

Each mm cohort follows its own non-parametric time trend. Units are compared only to other units within the same mm cohort.

More flexible than a linear trend control; fully non-parametric within cohort. Accounts for the sequential structure of the design.

What you are buying: parallel trends (assumption 2) need hold only among states with the same baseline dose *and* mm cohort (companion, assumption 5, p. 20); a switcher with no not-yet-switcher in its cohort is dropped.

Case 5 — The Precision Cost of Clean Identification

Why are CIs so wide for rm?

- dCH uses only *active switchers* for identification
- For rm: only 7 states switch $0 \to 1$ during 2002–2016, and the count falls with the horizon: **7 states at $\ell = 1$, 5 at $\ell = 2$, 2 at $\ell = 3$**
- Small effective N $\Rightarrow$ large standard errors
- Joint $p = 0.066$; only Effect_1 is significant at 5%, Effects_2 and $_3$ are not

What TWFE does instead:

Uses all mm=1 states, including earlier rm adopters, as controls: smaller SEs, but centred on a coefficient biased by heterogeneous effects.

Precision reflects the data

dCH identifies only from genuine variation, so it reports the precision the data actually support.

Wide CIs here mean too few switchers to be precise, not a zero effect.

Report the CIs; do not read wide intervals as "no effect."

Case 3 — East et al. (2023): Context & Data

Research question
Does prenatal Medicaid eligibility improve infant health (birth weight)?

Paper
East, Miller, Page & Wherry (2023), *AER* 113(1): 98–135.

Data (replication package)
50 states $\times$ **14 birth cohorts** (1975–1988); 700 obs.

Outcome `lbw_detrend81` (% low birth weight, detrended to 1981); treatment `newsimeli` (Medicaid eligibility rate).

Design (read off the data)

- **Continuous** treatment: 78 distinct values, range 0.04–0.28.
- **Absorbing**: eligibility only rises (28 increases, 0 decreases).
- **28** switchers.

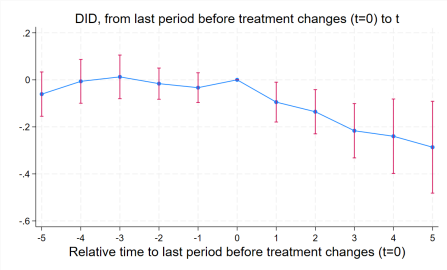
Use `continuous(1)`. The raw `Effect_`ℓ is then the **level** effect of having expanded eligibility (the numerator uses a binarised switch indicator); `Av_tot_eff` and `normalized` divide by the actual change in `newsimeli` and are **per unit of eligibility** (companion, pp. 22–23): a scaled level, not a derivative.

Case 3 — Continuous Treatment Results

```
* $stcontrols: state x year
*   demographics (see do-file)
did_multiplegt_dyn lbw_detrend81       ///
  plborn dob_y_p newsimeli,            ///
  effects(5) placebo(5) continuous(1)  ///
  controls($stcontrols) weight(births) ///
  effects_equal("all")
```

Results:

- Effects: $-0.095 \rightarrow -0.287$ (negative and growing); joint $p = 0.014$ ✓
- effects_equal $p = 0.186$ (stable); placebo $p = 0.721$ ✓
- 28 switchers



SEs shown are analytic; with `continuous()` dCH recommend `bootstrap`.

Reading

The low-birth-weight rate is 0.09 points lower in the year of an expansion, of 7.3 points of eligibility on average, and 0.29 points lower in the fifth year, four years later; per unit of eligibility (100 points) in force for one year, 2.67 points lower in total (`Av_tot_eff`), that is 0.27 per 10 points.

La tasa de bajo peso es 0.09 puntos menor el año de una expansión, de 7.3 puntos de elegibilidad de media, y 0.29 puntos menor en el quinto año, cuatro después; por unidad de elegibilidad (100 puntos) en vigor un año, 2.67 puntos menos en total, es decir 0.27 por cada 10 puntos.

Case 3 — The `effects_equal` Test

Null: $\text{Effect}_1 = \text{Effect}_2 = \cdots = \text{Effect}_5$. **East result:** $p = 0.186$, cannot reject.

What "not rejected" means here:

- Point estimates grow ($-0.095 \to -0.287$) but confidence intervals overlap
- With only 28 switchers, the test has limited power to detect heterogeneity
- Consistent with effects that build up and then stay stable

Av_tot_eff $= -2.67$: the *cumulative* effect *per unit of dose*, not a mean of the effects. A full unit dwarfs the changes (0.007–0.18): the level effects of the average 7.3-point expansion are -0.1 to -0.3; per unit (`normalized`), -1.3 to -0.8.

When to use Av_tot_eff

If effects_equal not rejected ($p \geq 0.05$): effect is stable. Report Av_tot_eff as the main result. The event-study is still shown for transparency.

If effects_equal rejected ($p < 0.05$): heterogeneous treatment effects by duration. Report each Effect_ℓ separately: the timing of the effect matters.

Inference under `continuous()`
The analytic SEs are conservative when the polynomial order is correctly specified, and can be liberal when it is set too high. Keep the order as low as the design allows, and report `bootstrap` SEs.

Option Selection Guide

Treatment type	dCH option	Example paper
Binary absorbing	use C&S instead	Most standard DiDs
Binary non-absorbing	base command only	Deryugina (hurricanes)
Count / intensity	`normalized`	Favara (banking laws)
Count + path check	`normalized` + `normalized_weights`	Gentzkow (newspapers)
Continuous	`continuous(1)`	East (Medicaid %)
Sequential binary	separate samples + `trends_nonparam`	Hollingsworth (mm/rm)

Always: `placebo(3+)` · `cluster()` at treatment level · report joint placebo p

Inference: analytic SE (conservative); bootstrap the command when `continuous()` is used

When Diagnostics Fail — The Practical Answers

The placebos reject. Now what?

In order: characteristic × year controls, `trends_lin`, `trends_nonparam()`. If none of them fixes it, the design does not hold and no option will rescue it. Deryugina is the honest example: nothing fixed it.

A rejection still tells you something useful. If the pre-switch gap in trends continues after the switch, the bias runs *opposite* to the placebo, so a negative placebo suggests your effect is overstated.

The interval is enormous.

Count the switchers per horizon first. Hollingsworth's recreational estimate rests on 7, 5 and 2 states. The width of that interval is a fact about the sample size.

The command refuses to run
If every group starts from a different period-one dose, it *needs* `continuous()`: no switcher has a same-baseline control, so failing without it is by design. Options that narrow the controls (`trends_nonparam()`, `only_never_switchers`) can drop switchers, which costs precision.

Standard errors
Analytic by default. With `continuous()` they lack proven theory: simulations suggest conservative at the right order, liberal when too high, so keep the order low and check against a bootstrap, which can crash on horizons with few switchers.

Report both the specification you defend and the one that made you doubt it.

Appendix — Advanced Options

Option	What it does
`switchers(in/out)`	estimate switchers-in and switchers-out separately (asymmetric effects)
`same_switchers`	hold the switcher sample fixed across all horizons (composition)
`trends_lin`	allow group-specific linear trends (no $\hat{\delta}$; one fewer placebo)
`dont_drop_larger_lower`	keep cells that cross the baseline (off by default)
`by()`	estimate separately by a group-level, time-invariant variable
`predict_het()`	test whether group-level, time-invariant covariates predict the estimated effects (not with `normalized`/`controls`)
`more_granular_demeaning`	possibly less-conservative SE if D changes more than once (large N)
`design`, `by_path`	`design`: list paths; `by_path`: effects by path

Full documentation: de Chaisemartin, Ciccia, Knau et al. (2025), arXiv:2510.19426.

Continuous Treatment: Callaway, Goodman-Bacon & Sant'Anna

From Case 3 to a Continuous Dose

Case 3 (East et al.'s data) gave us an effect per unit of eligibility under one assumption: with `continuous(1)`, that the status-quo trend is linear in the baseline dose. We assumed a functional form and obtained a scaled level.

That is one route. The question underneath it is more general, and continuous doses are everywhere: Medicaid eligibility rates, minimum wages in euros, tariff schedules, pollution exposure, subsidy shares.

The question for this module
With a continuous dose, what does parallel trends identify, and what does it fail to identify even when it holds exactly?

Callaway, Goodman-Bacon & Sant'Anna, *Difference-in-Differences with a Continuous Treatment*, forthcoming *AER*.

Did the Subsidy Work?

A subsidy covering a share d of a firm's costs. Suppose we knew the true effect at every level of coverage. Illustrative numbers.

Coverage d	Effect vs no subsidy
25 %	+2.0
50 %	+3.5
75 %	+4.4
100 %	+4.2

Each row compares one group of firms against firms that received nothing. Four separate comparisons, four heights above zero.

Every level of coverage beats having no subsidy, and by a wide margin. At full coverage the programme is worth +4.2.

If the question is whether to run it
This table settles it. The answer is yes, at every level of coverage.

Should We Raise It?

Same programme, same numbers, read by steps instead of by rows.

Step	Extra effect	Extra coverage	Per unit of coverage
25 % → 50 %	+1.5	0.25	$+1.5/0.25 = +6.0$
50 % → 75 %	+0.9	0.25	$+0.9/0.25 = +3.6$
75 % → 100 %	−0.2	0.25	$-0.2/0.25 = -0.8$

The last stretch of coverage made things worse: going from 75 % to full took the effect from +4.4 down to +4.2.

The programme is worth running at every level of coverage. The last stretch of it is not worth paying for.

Why the last column divides
Dividing by the step puts all three rows on the same footing: what a whole unit of coverage would add at that point on the curve. That is the number software reports, and it is why the figure changes if you measure the dose from 0 to 100 instead of from 0 to 1.

Two Causal Objects: Height and Slope

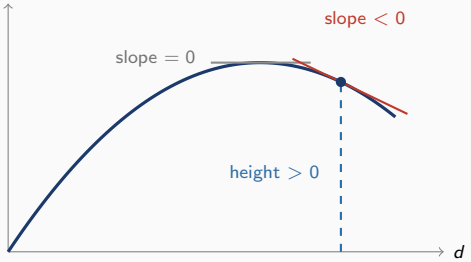
The two questions have names, and they are different parameters.

Level: $ATT(d \mid d)$
The effect of dose d against no treatment at all.
The *height* of the curve at d.

$$\mathbb{E}[Y_{t=2}(d) - Y_{t=2}(0) \mid D = d]$$

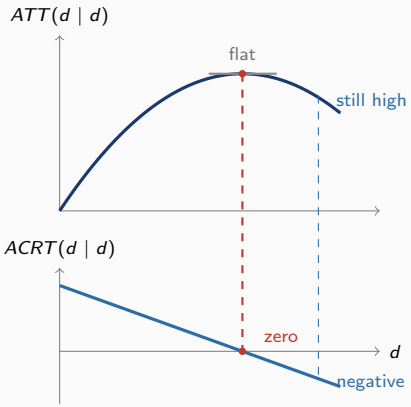
Slope: $ACRT(d \mid d)$
The effect of one more unit of dose. The *steepness* of the curve at d.

$$\mathbb{E}\Big[\tfrac{\partial}{\partial l} Y_{t=2}(l)\Big|_{l=d} \,\Big|\, D = d\Big]$$



A curve can be high and falling at the same time: ATT stays positive at every dose while $ACRT$ turns negative at the top. With a binary treatment there is one dose and one step, so the two coincide. The figure draws a single curve; whose curve it is comes after the reading of the two plots.

Reading the Two Plots

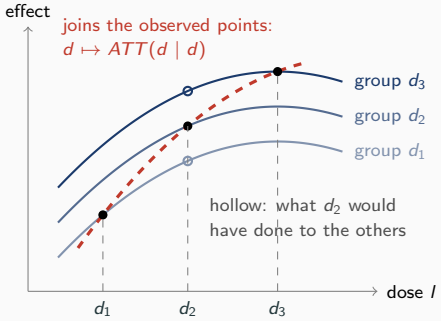


The command draws both. The second is the derivative of the first, redrawn so that a slope becomes a height.

- The **height** of the top plot at d is the level, $ATT(d \mid d)$.
- The **slope** of the top plot at d is the **height** of the bottom plot at d.
- Where the top curve flattens, the bottom one crosses zero.
- Where the top curve turns down, the bottom one is negative, and the level up there is still well above zero.

Why draw the second at all: you cannot put a confidence band on "that curve looks steep". Turning the slope into a height gives it a standard error.

One Point per Dose Group



- Each dose group has its own response curve: what every dose l would have done to it. Groups can respond by different amounts.
- The data show **one point per group**: the group at d_2 is observed at d_2 and nowhere else.
- **Local**, $ATT(d \mid d)$: the black dot of group d. ATT^{loc} averages the dots.
- **Global**, $ATT(d)$: the effect of dose d on all treated groups. At d_2, the vertical average of the black dot and the two hollow ones, which are never observed.

The red curve

It joins the black dots and is the only curve the data can draw. It is not the curve of any group: each point comes from a different group. Here it is steeper than every group's curve, because higher-dose groups respond more at every dose.

What Parallel Trends Identifies

What you can identify: the height of the curve
Under parallel trends, for every dose d in the support (Theorem 3.1, p. 10):

$$ATT(d \mid d) \;=\; \mathbb{E}[\Delta Y \mid D = d] - \mathbb{E}[\Delta Y \mid D = 0]\,,$$

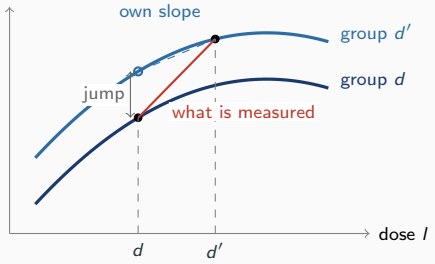
$$ATT^{\text{loc}} \;=\; \mathbb{E}[\Delta Y \mid D > 0] - \mathbb{E}[\Delta Y \mid D = 0]\,.$$

One DiD per dose, each against the untreated. Nothing exotic: it is the comparison of means you already know, run separately at every value of d.

- The whole curve $d \mapsto ATT(d \mid d)$ is identified.
- ATT^{loc} averages it over the treated, and equals a plain binary DiD on $\mathbf{1}\{D > 0\}$.
- Every comparison here is anchored on the untreated group.

In the picture: parallel trends pins the height of every black dot, and joined they trace the red curve. It does not justify comparing two dots with each other.

Differentiating the Joined Curve Compares Two Groups



Two black dots, one per group. The red chord between them is what a difference, or a derivative, measures, and it adds two pieces (Theorem 3.2, p. 11); divided by $d' - d$, the first is $ACRT(d' \mid d')$:

$$\underbrace{\text{own step of group } d'}_{ATT(d'|d')-ATT(d|d')} + \underbrace{\text{jump between curves at } d}_{ATT(d|d')-ATT(d|d)}$$

- The jump runs up to the hollow dot, what dose d would have done to group d'. Never observed, so **not identified**.
- Parallel trends says nothing about it. The paper calls it a version of **selection on gains** (p. 11).
- An infinitesimal step changes nothing: d against $d + \varepsilon$ is still two groups.

In the continuous limit: $\dfrac{\partial\, \mathbb{E}[\Delta Y \mid D = d]}{\partial d} = ACRT(d \mid d) + \left.\dfrac{\partial\, ATT(d \mid l)}{\partial l}\right|_{l=d}$.

Same Slopes, Still Biased

Two firms with the *same slope* from 25 % to 50 %: A receives 25 % coverage, B receives 50 %.

	Effect if coverage 25 %	Effect if coverage 50 %
Firm A (receives 25 %)	**2.0** (observed)	3.0
Firm B (receives 50 %)	3.0	**4.0** (observed)

What is measured
$4.0 - 2.0 = 2.0$. The causal response of going from 25 % to 50 % is 1.0 on both curves. The other 1.0 is that B responds one unit more than A at 25 % coverage.

What this says
The jump is the gap between the curves at the lower dose: $3.0 - 2.0$ at 25 %. Both curves start at zero effect at zero coverage, so that gap is the slope difference below 25 % (B rises 3.0, A rises 2.0). Equal slopes from 25 % to 50 % do not remove it.

Selection on gains: groups with different doses would gain different amounts from the *same* dose (here B gains more; in other data the gap goes the other way). Units may choose their dose for that reason, or exposure may go with responsiveness (for exposure doses the paper suggests "heterogeneity bias", fn. 2).

Strong Parallel Trends, and Why No Pre-Test Finds It

Strong parallel trends: at every dose d, all treated units, had they received d, would have followed the path dose group d actually followed (p. 12). With PT, **each black dot equals the average of the dots at its dose**; all groups sharing one curve is the simplest case.

What follows (Theorem 3.3, p. 13)
The joined red curve is then the global curve $ATT(d)$, the average over all treated. Its derivative is the $ACRT$, its maximum is the dose past which more adds nothing. With PT, $ATT(d) = ATT(d \mid d)$.

$$ACRT(d) = \frac{\partial \mathbb{E}[\Delta Y \mid D = d]}{\partial d}.$$

The paper, p. 13
"Strong parallel trends only changes the interpretation of the estimand, not its form."

The formula under PT and under SPT is **the same formula**: estimate the red curve and differentiate. What changes is whether the result may be called a causal response.

A pre-period only shows untreated potential outcomes, so no event-study plot, placebo or joint test can tell the two assumptions apart. The authors (p. 3): it is common for papers to write as if they had assumed the weak version and interpret as if they had assumed the strong one.

Four Summaries, Two Labels

	Level	Slope
Local: each group at its own dose	$ATT^{\text{loc}} = \mathbb{E}[ATT(D \mid D) \mid D > 0]$, the average of the black dots	$ACRT^{\text{loc}} = \mathbb{E}[ACRT(D \mid D) \mid D > 0]$, the average own-curve slope
Global: one dose for every treated group	$ATT^{\text{glob}} = \mathbb{E}[ATT(D) \mid D > 0]$, hollow dots included	$ACRT^{\text{glob}} = \mathbb{E}[ACRT(D) \mid D > 0]$

Firms: $ATT^{\text{loc}} = (2.0 + 4.0)/2 = 3.0$ needs only what is observed; $ATT(25\,\%) = (2.0 + 3.0)/2 = 2.5$ needs B's unobserved 3.0.

Parallel trends identifies one cell
The local level: ATT^{loc} is the binary DiD on $\mathbf{1}\{D > 0\}$. The global level needs the hollow dots. No slope exists with one point per curve, not even the local one.

Strong parallel trends fills the global row
With PT, $ATT(d \mid d) = ATT(d)$: local = global for the level. The joined curve's derivative is the global $ACRT(d)$, not $ACRT(d \mid d)$.

Which assumption gives which summary. PT identifies the local level ATT^{loc} (Thm 3.1); with SPT the same number is also global. The slope needs SPT, and the paper's SPT, which holds on average over dose groups, gives the global $ACRT^{\text{glob}}$ (Cor. 3.1). $ACRT^{\text{loc}}$ needs SPT for every dose group (App. C, fn. 13).

Software: the `contdid` Package

```
install.packages("contdid")   # R, CRAN 0.1.1
cont_did(yname = "Y", tname = "time_period", idname = "id", dname = "D",
         gname = "G", data = df,
         target_parameter = "slope",      # "level" for ATT(d|d)
         aggregation = "dose",            # or "eventstudy"
         control_group = "notyettreated",
         num_knots = 1, degree = 3, biters = 100, cband = TRUE)
```

The dose variable
It must be time-invariant, and it must carry its actual value in the pre-treatment periods. Do not code it as zero before treatment.

Not yet supported, per the authors
Discrete treatments, repeated cross-sections, unbalanced panels, time-varying doses, covariates. The data-driven dose model (`cck`) needs exactly two periods, so no staggered adoption.

The README calls it an alpha version and says the API may change.

Demo (2): A Hospital Panel With a Real Dose

The reform. Medicare's Prospective Payment System, 1983–84. Medicare had reimbursed a share of labor and capital costs; PPS replaced only the labor part, with a fixed fee per diagnosis, so capital became cheaper relative to labor.

Why a dose and not a switch. The reform is identical for every hospital. What differs is **exposure**: a hospital with 60 % of its patients on Medicare feels it far more than one with 10 %.

The dose. Medicare share of inpatient days in 1983; zero for hospitals with no Medicare patients (in the paper's data, children's and psychiatric).

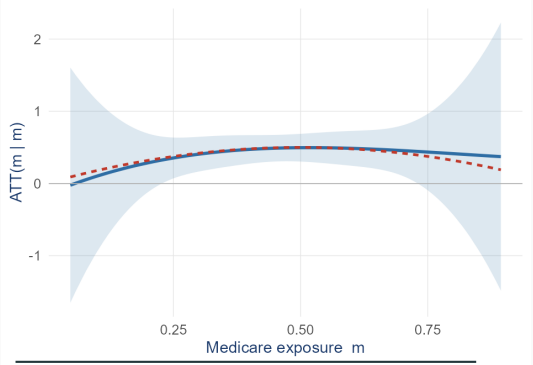
The outcome. Depreciation share of costs, a proxy for capital intensity.

The panel
6,000 hospitals, 1980–1986. **29 %** have zero exposure and act as the comparison group; the rest are spread over [0.05, 0.89]. The reform lands in 1984.

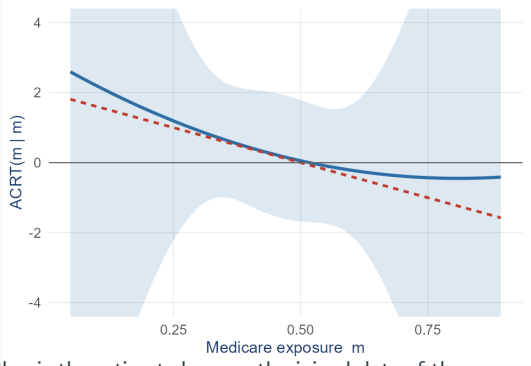
Pre- and post-periods are averaged into two, following the paper's own application (1980–83 against 1984–86, p. 25).

These data are simulated
The real AHA hospital survey must be bought from the association, so nobody can ship it. This is a third party's simulation of that panel, published in another paper's replication package. **It is not a replication of Acemoglu & Finkelstein, nor of the paper's application.** Its virtue is that the effect was imposed, so we know the answer.

Demo (2): Level and Slope Estimates



	Estimated	True
ATT^{loc}	0.4526	0.4548
$ACRT^{\text{glob}}$	0.3118	0.1713
Slope crosses zero at	$m = 0.517$	$m = 0.500$



Blue is the estimated curve, the joined dots of the theory, with its band; red dashed is the truth built into the simulation, one curve for every hospital. The true level is positive at every exposure. The slope changes sign near the middle, and past that point more exposure made things worse, where **38 %** of hospitals sit.

The slope summary is 0.68 SE from the truth. It looks further because $ACRT$ runs from $+1.8$ to -1.6 here, so the average cancels.

dCH `continuous()` and CGBS Compared

	dCH `continuous(K)`	CGBS
Assumption	status-quo trend is a degree-K polynomial in the baseline dose	parallel trends for the level, strong parallel trends for the slope
Identifies	the level effect of the switch by default; **per unit of dose** with `normalized` and in `Av_tot_eff`, a scaled level, not a derivative	$ATT(d \mid d)$ under PT, the global $ACRT(d)$ only under SPT
Control group	stayers, after netting out the modelled trend	the untreated, dose by dose
Non-absorbing	yes	no
Inference	bootstrap recommended	multiplier; opt-in uniform bands
Software	Stata, mature	R, alpha

The verdict of the JEL practitioners' guide (Baker et al. 2025, p. 61)
"These papers target different causal parameters, put more emphasis on different DiD designs, and, therefore, should be viewed as **complements rather than substitutes**."

dCH agree in print: their own survey lists CGBS as an estimator for the continuous-and-staggered design (2023, *Econometrics Journal*, Table 2, p. C27). If your treatment switches on and off, dCH is the one that applies. If it

Where We Have Got To

de Chaisemartin & D'Haultfœuille

- The control group is a set of **stayers with the same baseline treatment**.
- Identification comes from **many units switching at different dates**.
- Parallel trends is imposed on the **status-quo** outcome, and the placebos test it.

What changes now

- **One** treated unit, or a handful.
- No cohorts, no group-time effects, no asymptotics in the number of treated units.
- The control group has to be **built**, and its credential is the fit before treatment.

The question stays the same throughout: **who is the control, and what exactly is being compared?**

Synthetic Control & Synthetic DiD

Where We Are

Blocks 1 and 2 solved the multi-period, multi-cohort case:

- C&S: staggered binary absorbing treatment, many treated units
- dCH: non-binary, non-absorbing treatment; CGBS: continuous dose, staggered adoption from zero
- Key engine in both: *many* treated units $\rightarrow$ variation identifies the ATT

A case they cannot handle:

One (or very few) treated units

- A country adopts a policy. One region gets a shock. One city implements a program.
- With a single treated unit there are no cohorts, no group-time ATTs, and standard errors are not defined in the usual sense.

This block:

1. **Synthetic Control Method** (Abadie, Diamond & Hainmueller 2010): one treated unit, build a counterfactual from donor pool.
2. **Synthetic DiD** (Arkhangelsky et al. 2021): SCM unit weights + unit FE + time weights.

The Single-Unit Problem

Classic DiD compares treatment and control groups under parallel trends. With a single treated unit (a country, a region, a city) that logic strains:

- There is no treatment-group average, just one trajectory.
- The donor pool's *average* path need not parallel the treated unit's pre-treatment path.
- Sampling-based inference is hard to justify: no well-defined sampling mechanism, and often the sample is the population.

Why not just run DiD? DiD needs the treated–control gap to stay constant without treatment (parallel trends). For West Germany against the OECD average it already drifts before 1990 (Abadie 2021, Fig. 1A).

Synthetic control weakens it: rather than assume a comparable control exists, it *builds* one: a weighted combination of donor units that matches the treated unit's pre-treatment path.

Synthetic Control: The Idea

Core question: what would the treated unit's outcome have been *without* the intervention?

Synthetic control answer: find a weighted combination of donor units that *matches the treated unit's pre-intervention trajectory*; read the post-intervention divergence as the effect.

Intuition:

- No single donor looks like the treated unit.
- But a *convex combination* of donors often can.
- Weights are non-negative and sum to one, an interpretable average with no extrapolation.
- Post-intervention divergence $\approx$ the effect.

Key advantage

A data-driven procedure, not an informal choice, sets the donor weights, and the pre-period fit is shown, not hidden as in a regression. The analyst still chooses the donor pool.

The Setting

Abadie (2021, *Journal of Economic Literature*). One unit receives the intervention; the others do not.

$j = 1, \ldots, J+1$	the units. Unit 1 is the treated one; units $2, \ldots, J+1$ are the **donor pool**: untreated, and not affected by the intervention.
$t = 1, \ldots, T$	the periods; the first T_0 come before the intervention.
Y_{jt}	the outcome observed for unit j in period t.
Y_{jt}^N	the outcome unit j would have had with **no** intervention.
Y_{1t}^I	the outcome of the treated unit **with** the intervention.
$\tau_{1t} = Y_{1t}^I - Y_{1t}^N$	the effect on the treated unit in period $t > T_0$: one number per period, so the effect may build up or fade.

After T_0 the treated unit is observed with the intervention, $Y_{1t} = Y_{1t}^I$, so the missing piece is Y_{1t}^N: what would have happened without it. Estimating the effect is estimating that one counterfactual path. Every assumption below is about Y^N; nothing is assumed about Y^I.

The Synthetic Control

A synthetic control is a weighted average of the donors. The weights form a vector $\mathbf{W} = (w_2, \ldots, w_{J+1})'$ with

$$w_j \geq 0, \qquad \sum_{j=2}^{J+1} w_j = 1.$$

w_j is the share of the counterfactual that donor j contributes; the shares add up to one, so the counterfactual is one whole unit assembled from pieces of untreated units, none subtracted. West Germany in Abadie (2021): 0.42 Austria, 0.22 United States, 0.16 Japan, 0.11 Switzerland, 0.09 Netherlands. Given $\mathbf{W}$, the counterfactual and the effect are

$$\hat{Y}^N_{1t} = \sum_{j=2}^{J+1} w_j \, Y_{jt}, \qquad \hat{\tau}_{1t} = Y_{1t} - \hat{Y}^N_{1t}, \quad t > T_0.$$

Why this form. The comparisons everyone already makes are a particular $\mathbf{W}$: the plain donor average of a DiD ($w_j = 1/J$), a population-weighted average, or one nearest neighbour ($w_m = 1$). The method keeps the weighted average and lets the data set the weights.

The two constraints. Non-negative weights that sum to one keep the counterfactual inside the range of the donors: no extrapolation. The sum to one only makes sense if the variables are per capita or do not scale with size, such as prices.

The Predictors

For each unit, k predictors of its outcome that the intervention does not affect form a vector $\mathbf{X}_j$ ($k \times 1$): covariates, often averaged over pre-intervention years, and pre-intervention outcome values. $\mathbf{X}_1$ is the treated unit's column; $\mathbf{X}_0 = [\mathbf{X}_2 \cdots \mathbf{X}_{J+1}]$ ($k \times J$) holds one column per donor. West Germany: $k = 6$, the six rows of the table, one of them the outcome itself, GDP per capita averaged over 1981–90.

West Germany and the OECD average, Abadie (2021):

	West Germany	OECD average
GDP per capita	15,808.9	13,669.4
trade openness	56.8	59.8
inflation rate	2.6	7.6
industry share	34.5	33.8
schooling	55.5	38.7
investment rate	27.0	25.9

The plain donor average misses badly on GDP per capita, inflation and schooling.

Two ways to fill X. Some covariates plus a few outcome years, as here; or the whole outcome path before T_0, in which case any covariates added alongside stop influencing the weights (Kaul, Klößner, Pfeifer and Schieler 2022). A covariate that matters and is left out is not asked of the synthetic unit, which may then differ on it.

Choosing W: the Inner Problem

In practice the user supplies the predictors and the pre-intervention years, and the software returns $\mathbf{W}^*$. What it minimises:

$$\sum_{h=1}^{k} v_h \Big(X_{h1} - \sum_{j=2}^{J+1} w_j X_{hj} \Big)^2 \qquad \text{subject to } w_j \geq 0, \ \sum_j w_j = 1.$$

Term by term: for predictor h, the gap between the treated unit's value X_{h1} and the synthetic unit's $\sum_j w_j X_{hj}$, squared, times its importance v_h. $\mathbf{W}^*$ makes the sum of the k weighted gaps as small as possible, with non-negative weights that sum to one: the synthetic unit that resembles the treated unit most on the predictors, counting the important ones more. The v_h can be fixed by a rule (inverse variance) or chosen from the data in a second, outer problem.

Predictor	West Germany	synthetic
GDP per capita	15,808.9	15,802.2
trade openness	56.8	56.9
inflation rate	2.6	3.5
industry share	34.5	34.4
schooling	55.5	55.2
investment rate	27.0	27.0

Five of six predictors reproduced. Inflation is the exception: West Germany's 2.6 was too low for the synthetic unit to reproduce (fn. 16). The gaps $\mathbf{X}_1 - \mathbf{X}_0\mathbf{W}^*$ must be checked and reported.

Choosing V: the Outer Problem

v_h should measure how much predictor h helps to predict the outcome without intervention after T_0. That outcome is never observed, but before T_0 it is, so the software calibrates the importances on the years before the intervention: it tries importance vectors $\mathbf{V}$; for each one it computes $\mathbf{W}(\mathbf{V})$ as on the previous frame; it measures how closely that synthetic unit follows the treated outcome before T_0; and it keeps the $\mathbf{V}$ that follows it best:

$$\mathbf{V}^* = \arg\min_{\mathbf{V}} \sum_{t \le T_0} \left(Y_{1t} - \sum_{j=2}^{J+1} w_j(\mathbf{V})\, Y_{jt} \right)^2,$$

the sum, over the years before T_0, of the synthetic unit's squared error. In words: the outcome before the intervention decides how much each predictor counts, $\mathbf{V}$; matching the predictors with those counts decides how much each donor counts, $\mathbf{W}$.

What the user decides. Which predictors enter, which outcome years, and over which pre-intervention years the fit is measured. The search itself comes built in, in a quick approximate version and in a full, slower one. An alternative chooses $\mathbf{V}$ out of sample, on a training and a later validation stretch of the pre-period.

Not unique. The $\mathbf{V}$ chosen out of sample may not be unique (fn. 7); the paper asks to show that the results are not overly sensitive to the choice of $\mathbf{V}$.

What the Fit Tells You

Two checks follow the estimation, and both are visible in the data.

1. **The predictors: $\mathbf{X}_1 \approx \mathbf{X}_0\mathbf{W}^*$**, predictor by predictor: the balance table of "Choosing $\mathbf{W}$".
2. **The outcome before T_0:** the synthetic unit should follow Y_{1t} over the whole pre-intervention period. In West Germany the plain OECD average drifts away from GDP per capita for decades before 1990; the synthetic unit tracks it until 1990. One number summarises this fit: the root mean squared prediction error before T_0, the typical distance between Y_{1t} and $\hat{Y}_{1t}^N$ in the outcome's units; the placebo test divides by it.

What the method exploits is the co-movement of the outcome across units: GDP per capita moves together across countries, so a weighted average of a few of them can follow one of them for thirty years.

Post-intervention outcomes never enter
The weights depend on pre-intervention data only. They can be computed, and registered, before the post-intervention outcomes exist.

Inference: the In-Space Placebo

In a nutshell. Give the intervention, on paper, to each donor in turn and estimate its "effect" with the same method. Nothing happened to the donors, so those effects are what the method produces by chance. The treated unit's effect is evidence when it is larger than all, or almost all, of them.

The p-value. The share of the $J+1$ units whose effect is at least as large as the treated unit's. If the treated unit ranks first, $p = 1/(J+1)$, and no smaller value exists: with 20 donors, p never goes below $1/21$.

How "large" is measured. By the gap after T_0 relative to the fit before T_0, both as root mean squared gaps, so that a donor the method never fitted well does not pass for an effect:

$$r_j = \frac{\text{root mean squared gap after } T_0}{\text{root mean squared gap before } T_0}.$$

In-Space Placebo: the Cutoff and the Three p-Values per Year

The cutoff. A donor whose synthetic control fits badly before T_0 has large gaps in every year, before and after, and looks "extreme" without any effect. Donors whose pre-period mean squared gap is more than k times the treated unit's are dropped before the year-by-year p-values; with m units left, the smallest p is $1/m$.

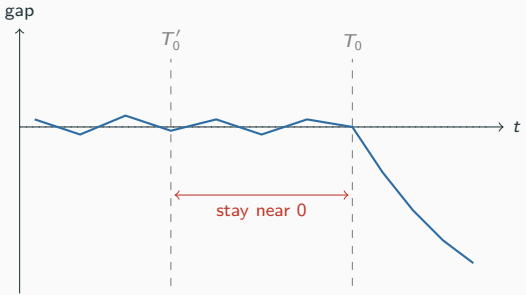
Three p-values per year. For each year after T_0, the treated unit's gap is compared with the placebo gaps of that same year, the treated unit counting itself:

- **left-sided:** the share of gaps at least as negative as the treated unit's;
- **right-sided:** the share at least as positive;
- **two-sided:** the share at least as large in absolute value, in either direction.

Six units, treated gap -10, placebo gaps -12, -3, 0, $+4$, $+11$: left-sided $2/6$ (-12 and the treated unit), right-sided $5/6$ (all but -12), two-sided $3/6$ (-12, $+11$ and the treated unit).

Robustness: the In-Time Placebo

Move the intervention back to a year $T_0' < T_0$ when nothing happened, drop every predictor measured after T_0', and re-estimate as if the intervention had come then.



Between T_0' and T_0 the gap should stay near zero: the synthetic unit keeps following the treated unit when nothing happens to it.

A gap that opens before T_0 is a drift of the method itself. The effect after T_0 then has to be much larger than that drift to count.

Robustness: Leave-One-Out and the Predictor Set

Leave-one-out. Drop each donor with positive weight, one at a time, and re-estimate. Trajectories that stay close to the original say the result rests on no single donor; one that jumps names the donor it rests on. Abadie (2021, Fig. 4) plots every trajectory beside the original; here we also report the range of the gap.

The predictor set. Another set of predictors gives another $\mathbf{W}^*$, another fit before T_0 and another gap. To choose the set, fit the weights on the earlier pre-years and keep the set that best predicts the later ones (p. 402). Show the alternatives beside it; the conclusion should hold across them (p. 415).

Requirements: Context and Data

Context

- **Effect against volatility.** A small effect is lost in the treated unit's own shocks; common shocks are differenced out, unit-specific ones are not.
- **A comparison group exists.** Drop donors with similar interventions or large idiosyncratic shocks; the rest should resemble the treated unit.
- **No anticipation.** If agents react early, move the intervention date back in the data.
- **No interference.** Drop exposed donors; same-sign spillovers left in attenuate the estimate.
- **Within reach of the donors.** Level out of reach and only the trend matters: differences, growth rates, demeaning or an intercept (synthetic DiD).
- **Time horizon.** Effects that build up need post-periods.

Data

- Aggregate series for predictors and outcomes, official or built from microdata.
- A long pre-intervention window: a short one invites a fit that is spurious. If the window is long enough to hold structural breaks, weight recent predictors more through $\mathbf{V}$.
- Enough post-intervention periods for the effect to appear.

Synthetic Control: Published Applications

The same recipe, one treated unit, a long pre-period and a credible donor pool, in five published cases:

- **Abadie & Gardeazabal (2003)**: the economic cost of terrorism in the Basque Country (the original SCM).
- **Abadie, Diamond & Hainmueller (2010)**: California's Proposition 99 tobacco tax *(our demo)*.
- **Abadie, Diamond & Hainmueller (2015)**: German reunification and per-capita GDP.
- **Born et al. (2019)**: the Brexit vote and UK output.
- **Grier & Maynard (2016)**: the economic consequences of Hugo Chávez.

Conflict, macro, trade, politics: the same method. The demo works the second case.

Demo: California Prop 99 — SCM

Case — California Prop 99: Context & Data

Research question

Did Proposition 99 (in force January 1989; cigarette tax raised by \$0.25/pack) cut per-capita tobacco consumption?

Paper / data

Abadie, Diamond & Hainmueller (2010), *JASA*; `smoking.dta`.

- Panel: **39 US states**, 1970–2000.
- Treated: **California** (`state==3`); donor pool: 38 states.
- Outcome `cigsale` (packs p.c.); predictors: log GDP p.c., beer, share aged 15–24, retail price, lagged `cigsale`.

Why SCM here?

One treated unit $\Rightarrow$ C&S / dCH do not apply. No single state matches California's pre-1989 path, but a convex combination of donors can.

We implement it with `synth2` (Yan & Chen 2023), which wraps `synth` and automates placebo tests, the leave-one-out check, and the figures.

Stata: the synth2 Command

```
use "$data/smoking.dta", clear
xtset state year                          // state 3 = California

synth2 cigsale lnincome age15to24 retprice beer ///
    cigsale(1988) cigsale(1980) cigsale(1975),  ///  predictors + lagged outcomes
    trunit(3) trperiod(1989)                    ///  treated unit & date
    xperiod(1980(1)1988) nested allopt              // ADH spec
```

- `trunit(3) trperiod(1989)`: California, treated from 1989.
- `xperiod(1980(1)1988)`: average the predictors without their own year over 1980–1988.
- `nested allopt`: the full search for **V**. Every block keeps it except the in-space placebo, where 39 full searches would take over an hour.
- Add `placebo(unit cutoff(2))`, `placebo(period(1985))`, or `loo` for inference and robustness (next slides).

Output: Synthetic California

Optimal unit weights, the $\mathbf{W}^*$ of "Choosing $\mathbf{W}$":

State	Weight
Utah	33.5%
Nevada	23.5%
Montana	20.2%
Colorado	16.0%
Connecticut	6.8%

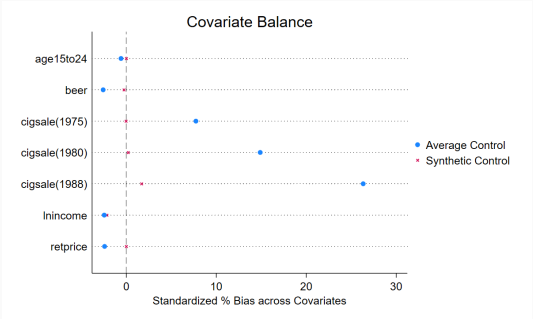
All other 33 donors get weight 0: with California outside the donors' convex hull, at most as many donors as predictors (7) carry weight (Abadie 2021).

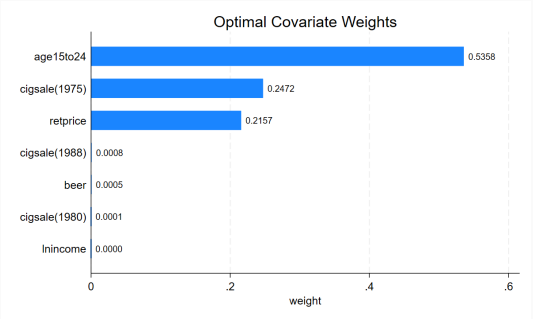
Pre-treatment fit:

- **RMSPE** $= 1.75$, $\quad R^2 = 0.97$.
- Predictor balance is tight: synthetic California matches the treated unit far better than the simple donor average (e.g. `cigsale(1988)` bias 1.7% vs 26%).
- Predictor importance ($\mathbf{V}$) concentrates on `age15to24`, `cigsale(1975)` and `retprice` (next frame).

Same five donors as Abadie et al. (2010, Table 2); `synth2`'s weights differ from theirs by at most 0.4 points.

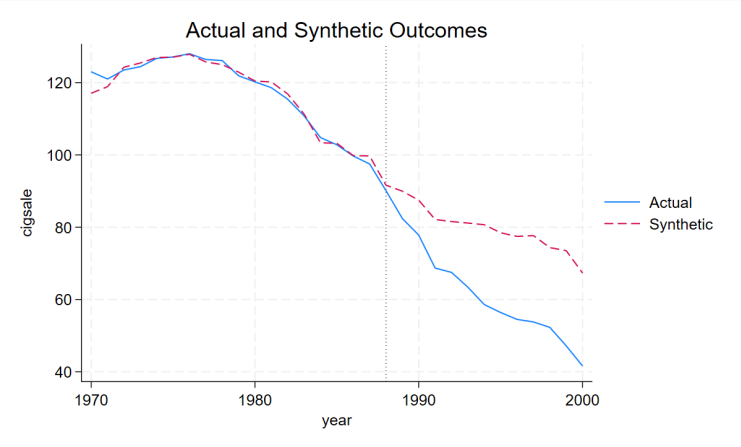
Output: Balance and Predictor Importance





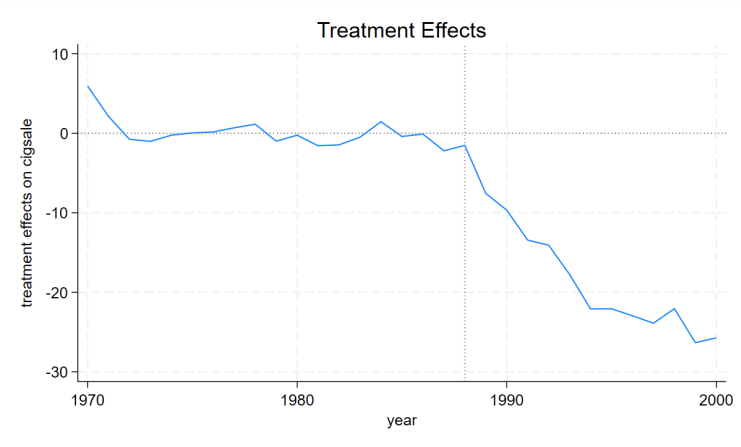
Left: predictor balance, the gaps $\mathbf{X}_1 - \mathbf{X}_0\mathbf{W}^*$ of the theory as a plot. For each predictor, how far California is from the synthetic unit built with $\mathbf{W}^*$ (crosses) and from the plain donor average (dots), in per cent of California's value. The synthetic unit is within 2.2 % on every predictor; the plain average is 26 % off on 1988 sales and 15 % on 1980 sales. **Right:** the importance v_h the search gave each predictor ("Choosing $\mathbf{V}$"). Three carry all of it: `age15to24` 0.54, `cigsale(1975)` 0.25, `retprice` 0.22. Other searches return other $\mathbf{V}$ and other weights: the quick search of the placebos gives RMSPE 2.04 and ATT -18.0 (here 1.75, -19.0).

California vs Synthetic California



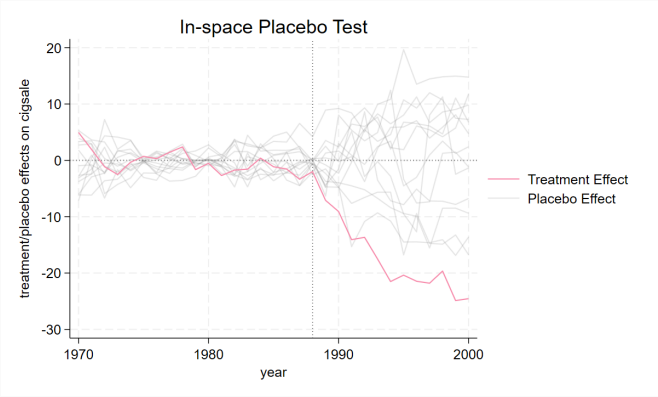
Reading it: the second check of "What the Fit Tells You": the two series track closely before 1989 (RMSPE 1.75 packs). After Prop 99, California diverges downward.

The Effect Over Time



Reading it: the effect $\hat{\tau}_{1t}$ of "The Synthetic Control", year by year. The gap is ≈ 0 pre-1989, then grows to ≈ -26 packs by 2000. Average post-treatment effect (ATT) $= \mathbf{-19.0}$ packs per capita.

Inference: In-Space Placebo Test

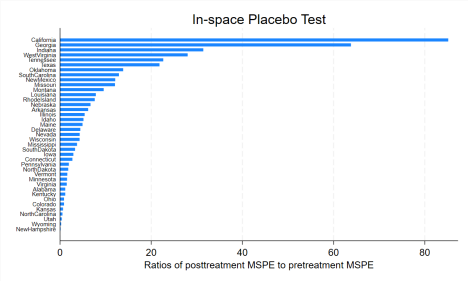


Each of the 39 units gets its own synthetic control; the figure shows California (pink) and the 13 donors kept by `cutoff(2)`. California's gap is the most extreme from 1992 on. This block runs the quick search, so California's own fit here is RMSPE 2.04 rather than 1.75.

One number per unit: the mean squared gap after 1989 divided by the mean squared gap before, the ratio the command prints (taking roots would not change the ranking). California 85; Georgia comes next with 64; 29 of the 38 donors are below 10.

The *p*-value: California ranks first of 39, so $p = 1/39 = 0.026$, the floor.

In-Space Placebo: the Ranking and the Cutoff



The 39 ratios ranked, one bar per unit: California 85, Georgia 64, Indiana 31, then a long tail. First of 39: $p = 1/39$.

Rows of the table the command prints (mean squared gaps):

unit	pre	post	ratio	pre / pre CA	`cutoff(2)`
California	4.2	353.6	85.1	1.0	
Georgia	1.9	122.8	63.8	0.5	kept
Indiana	40.6	1276.1	31.4	9.8	out
Nevada	59.6	256.1	4.3	14.3	out
Colorado	36.5	31.8	0.9	8.8	out
Connecticut	27.2	73.4	2.7	6.6	out
Utah	593.8	223.3	0.4	143.0	out
Kentucky	683.5	778.8	1.1	164.6	out
New Hampshire	3485.5	312.8	0.1	839.3	out

The cutoff: a donor stays if its pre-1989 error is at most twice California's, last column ≤ 2. 13 of the 38 donors stay; with California, 14 units. Utah, Nevada, Colorado and Connecticut, four of California's five donors, have no good synthetic control of their own and go; Montana stays. California still ranks first among the 14: $p = 1/14 = 0.071$, the floor for every year-by-year p.

In-Space Placebo: Three p-Values per Year

The table the command prints, among the 14 units kept:

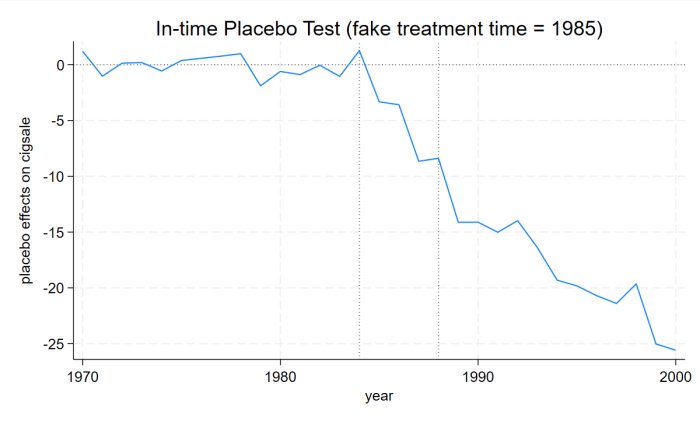
year	gap	two-sided	right	left
1989	−7.1	0.143	1.000	0.071
1990	−9.1	0.143	1.000	0.071
1991	−14.1	0.143	0.929	0.143
1992	−13.7	0.071	1.000	0.071
1993	−17.5	0.071	1.000	0.071
1994	−21.5	0.071	1.000	0.071
1995	−20.4	0.071	1.000	0.071
1996	−21.5	0.071	1.000	0.071
1997	−21.8	0.071	1.000	0.071
1998	−19.6	0.071	1.000	0.071
1999	−24.9	0.071	1.000	0.071
2000	−24.5	0.071	1.000	0.071

Each year, California's gap against the 13 placebo gaps of that year, California counting itself: $0.071 = 1/14$, $0.143 = 2/14$.

- **Left**, gaps at least as negative as California's: $1/14$ every year but 1991, when one donor fell more. The one to read: Prop 99 was meant to cut sales.
- **Right**, gaps at least as positive: $14/14$ every year but 1991. California's gap is the lowest, so every placebo gap lies above it.
- **Two-sided**, gaps at least as large in either direction: $2/14$ in 1989–1991, $1/14$ from 1992. In 1989–90 one donor had a positive gap as large as California's negative one.

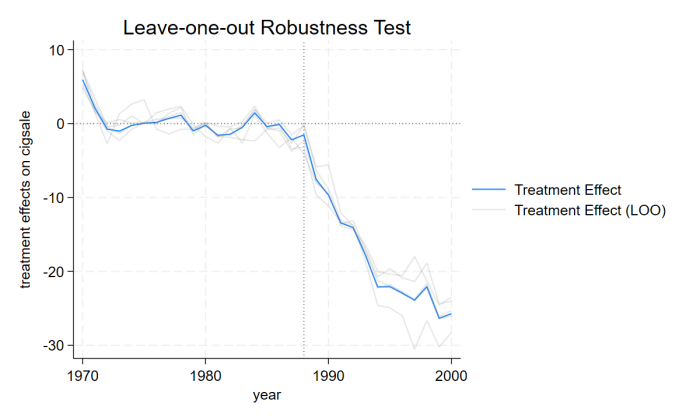
The three plots the command draws are these three columns against the year.

Robustness: In-Time Placebo (fake 1985)



Reading it: the check of "Robustness: the In-Time Placebo". With the treatment moved to 1985 the gap stays small in 1985–86 (≈ -3), but it already reaches ≈ -8 by 1987–88, a mild pre-existing drift. The *much* larger divergence (to -25) still comes after 1989, so the effect is not purely a pre-trend; but read this as a *weaker* check than the in-space test.

Robustness: Leave-One-Out



Reading it: the check of "Robustness: Leave-One-Out and the Predictor Set". Dropping each of the five weighted donors in turn (gray) barely moves the estimated effect (blue). By 2000 the effect is -25.7 with a leave-one-out range of $[-28.3, -23.5]$, so it is not driven by any single state.

Interpretation: SCM

Findings:

- Synthetic California (Utah, Nevada, Montana, Colorado, Connecticut) tracks real California closely before 1989.
- Gap grows to ≈ 26 fewer packs by 2000; ATT $= -19.0$ over 1989–2000 (21 % of the 1988 level of 90 packs).
- In-space placebo $p = 0.026$ (the strong evidence) and leave-one-out is stable; the in-time check is weaker (a mild pre-1989 drift).

Caveats:

- SCM gives a trajectory, not a single SE-bearing ATT; inference is permutation-based.
- Needs a good pre-fit (here RMSPE 1.75 vs a mean $\sim$116) and no spillovers to the donor pool.

Result
California's tax cut per-capita consumption by $\sim$19 packs/year on average (26 by 2000), a large and robust effect (placebo rank 1 of 39).

When Synthetic Control Should Not Be Used

Abadie (2021) sets out the conditions the method needs. Nine ways they fail, grouped by what goes wrong.

The fit is not there

- Poor pre-treatment fit. Abadie et al. (2010) "recommend against" it, and **a long pre-period cannot rescue a bad fit**.
- The treated unit's level is beyond any weighted average of donors, and the counterfactual needs the level, not only the trend: the estimator "should not be used in that case".
- Few pre-periods, which invites a fit that is spurious.

The comparison is not clean

- No uncontaminated donor pool.
- Anticipation you cannot date.
- Spillovers you can neither drop nor sign.
- Several treated units inside the donors' hull: exact fits from distant donors hide bias, an "illusion of perfect fit".

The signal is too small
An effect smaller than the unit's own idiosyncratic volatility, or a post-period too short to show it.

Synthetic DiD

From Synthetic Control to Synthetic DiD

Three ways to say what would have happened to the treated unit:

- **DiD:** it would have changed as the plain average of the controls changed. Every control counts the same, every pre-year counts the same. Needs parallel trends.
- **Synthetic control:** it would have followed a weighted average of donors chosen to reproduce its *level* before T_0. The effect is the gap after T_0.
- **Synthetic DiD** (Arkhangelsky, Athey, Hirshberg, Imbens and Wager 2021): a DiD in which the control is a weighted average of donors chosen to move *in parallel* with the treated unit before T_0, and "before" is a weighted average of the pre-years that look most like the post-period.

In words: synthetic DiD keeps the DiD comparison, change against change, and lets the data choose which controls and which pre-years to compare with. The level need not match, only the trend: the level gap is absorbed, as in any DiD.

Why one of the two sets of weights is enough
The bias is the product of two errors: how far the weighted donors are from parallel, times how badly the weighted pre-years predict the post-period. If either set of weights does its job, the product is small.

Synthetic DiD Step by Step

1. **Unit weights** ω_j, one per donor, non-negative, summing to one: chosen so that the weighted donor average runs parallel to the treated unit before T_0. An intercept absorbs the level difference, so a donor 20 units higher but moving in step is a good donor. A penalty spreads the weight over many donors instead of two or three.
2. **Time weights** λ_t, one per pre-year, non-negative, summing to one: chosen so that, for the donors, the weighted average of the pre-years predicts the post-period average up to a constant. Pre-years that look like the post-period count; years that look nothing like it get no weight.
3. **The estimate:** a DiD with those weights.

	before: λ-weighted pre-years	after: post-period average
treated unit	$\sum_t \lambda_t \, Y_{1t}$	$\bar{Y}_{1,\text{post}}$
donors, ω-weighted	$\sum_j \omega_j \sum_t \lambda_t \, Y_{jt}$	$\sum_j \omega_j \, \bar{Y}_{j,\text{post}}$

$\hat{\tau} = (\text{after} - \text{before})_{\text{treated}} - (\text{after} - \text{before})_{\text{donors}}$: the change of the treated unit minus the change of its weighted donors. With every weight equal it is the plain DiD; with no intercept, weights that match the level and no "before" column, it is the synthetic control.

What the Intercept, the Penalty and the Time Weights Do

The intercept. The weighted donors need only move in parallel with the treated unit; the level gap is absorbed. Synthetic control has no intercept and must match the level too, which rules out donors that are right on trend and off on level.

The penalty on the unit weights. Its size is a typical one-year change of an untreated unit, scaled up by $(N_{tr}\, T_{post})^{1/4}$ (about 1.9 for California). It spreads the weight over more donors, so no single one carries the result, and it makes the weights unique.

The time weights. Pre-years that look nothing like the post-period get no weight. DiD gives every pre-year the same weight, so a distant year in which the donors behaved differently from now pulls the counterfactual.

California, average over 1989–2000, from Arkhangelsky et al. (2021):

	effect	SE
DiD, uniform weights	−27.3	17.7
SC, unit weights	−19.6	9.9
SC + intercept	−11.1	9.5
SDID, + time weights	−15.6	8.4

The intercept moves synthetic control's −19.6 to −11.1; the time weights bring it to −15.6. The standard error falls at each step (9.9, 9.5, 8.4). When synthetic control and synthetic DiD disagree, the paper argues the latter is often more credible.

Synthetic DiD Over Time: the Event-Study Version

The single $\hat{\tau}$ averages the whole post-period. With the same weights one can estimate one effect per period since adoption, as in the event study of Block 2 (Ciccia 2024):

- **Effects** $\hat{\tau}_\ell$, $\ell = 1, 2, \ldots$: the DiD of "Synthetic DiD Step by Step" with "after" replaced by the ℓ-th period from adoption ($\ell = 1$ is the adoption period). They show whether the effect appears at once, builds up or fades.
- **Placebos** $\hat{\tau}_{-\ell}$, $\ell = 1, 2, \ldots$: the same DiD with "after" replaced by the ℓ-th period *before* adoption, an effect estimated where there was none. Placebos near zero say the weighted donors followed the treated unit before adoption; a placebo far from zero is a warning about the counterfactual.

With staggered adoption each cohort has its own weights and its own effects by period since adoption; at each ℓ the event study averages the cohorts, each weighted by its number of treated units. Inference as for the single $\hat{\tau}$: placebo with one treated unit, bootstrap with several.

Synthetic DiD Inference, in Words

The estimate is a weighted DiD, so it comes with a standard error. The paper gives three ways to get one:

- **Placebo.** Set the treated units aside; give the treatment, on paper, to as many controls as there were treated, chosen at random; re-estimate; repeat. The spread of the placebo estimates is the standard error. The only option with one treated unit, and valid only if all units share the same noise variance.
- **Bootstrap.** Resample units with replacement and re-estimate. Needs several treated units, so that every resample contains some.
- **Jackknife.** Drop one unit at a time and recompute the weighted DiD with the weights held fixed, so SDID runs once. Not defined with one treated unit.

What to report
The method, the seed and the number of replications. The point estimate is fixed; the placebo and bootstrap standard errors are simulated and move with the draw.

SDID vs SCM vs DiD: When to Use What

	DiD / TWFE	SCM	SDID
# treated units	Many	One (or few)	One or many
Before T_0	Parallel trends *assumed*	Level matched	Trend matched
Unit weights	Uniform	Few donors	More than SCM
Time weights	Uniform	None	Chosen by the data
Standard errors	Cluster SE	Permutation p	Placebo, jackknife, bootstrap
Multiple treated	Yes (C&S, dCH)	Yes (averages them)	Yes

Nesting: SDID contains both DiD (all weights equal) and SCM (unit weights matching the level, no unit fixed effects, no time weights) as special cases.

Staggered adoption: one block per adoption cohort, each with its own weights, then averaged (Clarke, Pailañir, Athey and Imbens 2024).

Demo: Synthetic DiD

Stata: the sdid Command (California)

```
use "$data/smoking.dta", clear
gen treated = (state == 3 & year >= 1989)   // California from 1989

sdid cigsale state year treated, ///
    vce(placebo) seed(1234) reps(100) graph
```

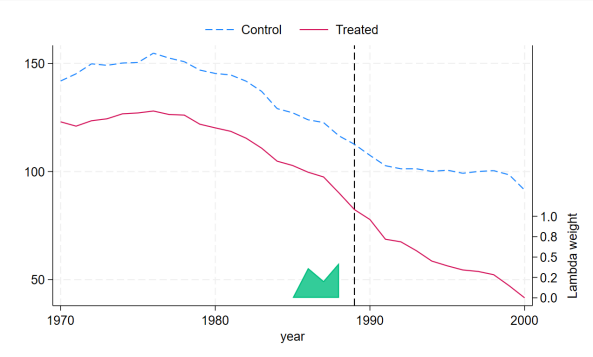
Syntax: `sdid Y unit time treated [, options]`

- `vce(placebo)`: placebo SEs: the mode for a single treated unit (seed + reps pinned for reproducibility).
- `graph`: reweighted control vs treated path, plus the $\hat{\lambda}_t$ time weights.

Result: ATT $= -\mathbf{15.60}$ packs (SE 8.99, $p = 0.082$, 95% CI $[-33.22,\ 2.01]$), with `seed(1234) reps(100)`.

Always report both. The point estimate is stable, the placebo SE is not: Clarke et al. (2024) get 9.879 and $p = 0.114$ with a different seed and `reps(50)`, and the original paper reports 8.4. With one treated unit the verdict crosses $p = 0.10$ on the luck of the draw.

SDID California: Weighted Trajectories + Time Weights



Reading it: blue dashed = unit-weighted control average; red = California; green = the time weights $\hat{\lambda}_t$, non-negative and summing to one over 1970–1988: 0.37 on 1986, 0.21 on 1987, 0.43 on 1988, zero on every earlier year. The unit weights $\hat{\omega}_j$ are positive for 28 of the 38 donors (largest 0.12), where the synthetic control used five. One ATT for 1989–2000 is reported; the yearly gaps with their bands are the event study two frames on.

Same Data, Three Estimators

California, our `sdid` output:

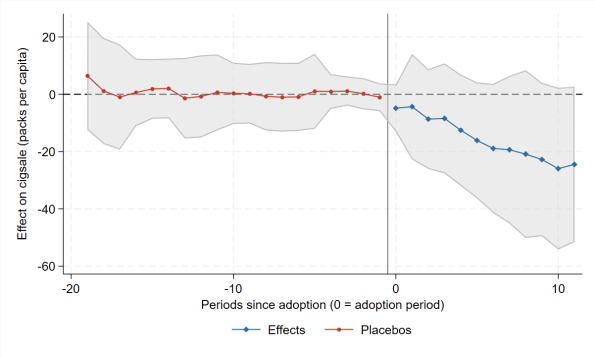
Estimator	ATT (packs)
DiD	-27.35
SC	-19.62
SDID	$\mathbf{-15.60}$

These reproduce Arkhangelsky et al. (2021) exactly ($-27.3/-19.6/-15.6$).

Why they differ:

- **DiD** uses all controls with equal weights, biased if pre-trends are not parallel (they are not here).
- **SC** reweights units to match the pre-period level; the 2000 gap alone is ≈ -26.
- **SDID** adds unit *and* time weights (the latter on pre-years only) and reports one ATT, the equal-weight average over 1989–2000 (-15.6 vs SC's -19.6), with a placebo SE that assumes homoskedasticity across states.

SDID Event Study: California




```
sdid_event cigsale state year treated, ///
    vce(placebo) brep(100) placebo(all)
```

`sdid_event` (Ciccia 2024): "Synthetic DiD Over Time" with the weights of the `sdid` run.

Effects (blue): -4.8 packs in 1989, -12.5 in 1993, -26 by 1999. Their average is the ATT of `sdid`, -15.6.

Placebos (red): between -1.4 and $+2.0$ packs in every pre-year but 1970 ($+6.5$).

Bands: every yearly band includes zero. One treated unit and 100 placebo draws give standard errors of 4 to 15 packs, and the ATT's band, $[-33.8, 2.6]$, includes zero too ($p \approx 0.09$). The verdict on California is the permutation rank of the synthetic control, first of 39; the event study adds the shape.

SDID with Staggered Adoption: Context & Data

Research question

Do reserved-seat gender quotas raise women's representation in parliament?

Paper / data

Clarke et al. (2024); Bhalotra, Clarke, Gomes & Venkataramani (2023). `quota_example.dta`.

- **119 countries**, yearly panel 1990–2015; 9 adopt a quota.
- Outcome `womparl` (% women in parliament); treatment `quota` (0/1).
- **Staggered**: quotas adopted in **7 different years** (2000–2013).

Why SDID here?

Multiple treated units adopting at different dates. `sdid` handles this by splitting the panel into adoption cohorts (one block each) and averaging, so the same engine extends to staggered designs.

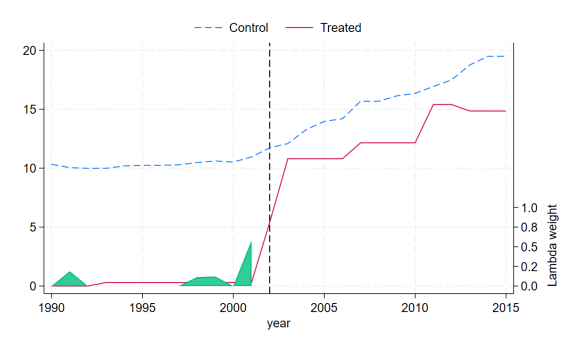
SDID Staggered: Estimation

```
use quota_example, clear
sdid womparl country year ///
    quota,                ///
    vce(bootstrap)        ///
    seed(1234) reps(100) graph
```

`vce(bootstrap)`: resamples units (valid with many treated).

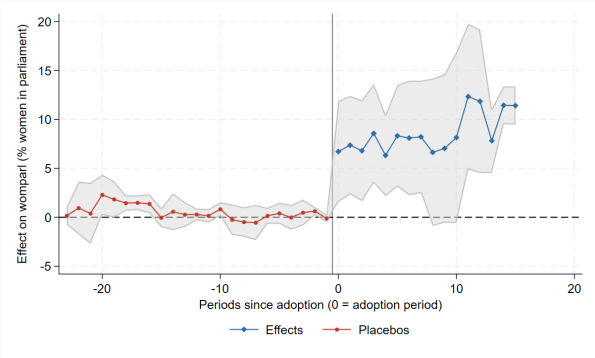
Result: ATT $= +\mathbf{8.03}$ pp
(SE 3.31, $p = 0.015$, CI [1.54, 14.53]).

Quotas raise women in parliament by ≈ 8 percentage points.



One adoption cohort (2002): treated countries jump after the quota; $\hat{\lambda}_t$ (green) weight the pre-2002 years.

SDID Event Study: Quotas, Staggered




```
sdid_event womparl country year quota, ///
    vce(bootstrap) brep(100) placebo(all)
```

One event study across the seven adoption cohorts (nine countries), each cohort with its own weights.

Effects: +6.7 points in the adoption year, 6 to 9 over the next ten years. Beyond that, 11 to 12 (7.8 thirteen years after adoption), resting on five, three and one countries: the jump is a change of composition.

Placebos: below one point up to fifteen years before adoption; 1.4 to 2.3 at 16 to 20 years back (five countries); below one again beyond.

ATT = 8.03, as with `sdid`; SE 2.49 here against 3.31 there, each from its own bootstrap of 100 replications.

SCM and SDID — Takeaways

Synthetic Control Method:

- Best for 1 treated unit, long pre-period and good pre-fit.
- Constructs counterfactual as a convex combination of donors.
- Inference via in-space permutation (placebo p-values; Fisher-exact only under random assignment).
- `synth2` in Stata.

Synthetic DiD:

- Adds unit fixed effects and time weights to SC-style unit weights.
- Unit weights make pre-trends parallel; time weights pick the pre-years that predict the post-period.
- A single ATT with a standard error, or one effect per period since adoption with its placebos.
- `sdid` and `sdid_event` in Stata.

Decision rule (thumb)

1 treated unit, trajectory + permutation p	$\rightarrow$ SCM
1+ treated units, single ATT + SE	$\rightarrow$ SDID
Staggered adoption	$\rightarrow$ SDID (cohorts) or C&S/dCH

SCM and SDID — The Practical Answers

How small a p can I get?

With 38 donors the permutation test cannot go below $1/39 = 0.026$, however large the effect. If a referee wants $p < 0.01$, permutation inference cannot give it. And `cutoff()` discards donors, so it moves the p by changing the denominator.

Which standard error?

With one treated unit, `vce(placebo)`; the command refuses the bootstrap there. Always report the seed and `reps()`: the point estimate is stable, the placebo SE is not.

How long a pre-period?

Long enough to fit, but length does not rescue a bad fit. A poor pre-period fit is a reason not to use the method at all.

What goes in the report

The path plot with the treated and synthetic series, the weight table so the reader can see which donors carry the estimate, the permutation p-value, and at least one of leave-one-out or the in-time placebo. The weights are what makes the exercise auditable by someone who is not an econometrician.

A single ATT can hide everything

In the staggered quota example the aggregate is $+8.03$ while the seven adoption cohorts run from -3.45 to $+21.76$. Always look at `e(tau)` before you quote the headline.

Who Is the Control: Five Methods

Method	Who is the control	What you must defend
dCH	Stayers with the *same* period-one treatment	Parallel trends on the status-quo outcome; no crossing
dCH `continuous()`	No two groups share a baseline, so the status-quo trend is modelled as a polynomial in it	The order of that polynomial
Callaway, Goodman-Bacon & Sant'Anna	Units with $D = 0$, dose by dose	Parallel trends for levels; *strong* parallel trends before you read any slope
Synthetic control	A convex combination of donors that reproduces the pre-period	Fit before treatment, and a donor pool that can produce it
Synthetic DiD	The same, reweighted over units *and* periods	The same, with inference that suits the number of treated units

No estimator removes the need to say which comparison you are making. Each of these makes the comparison explicit, which is what lets a reader disagree with it.